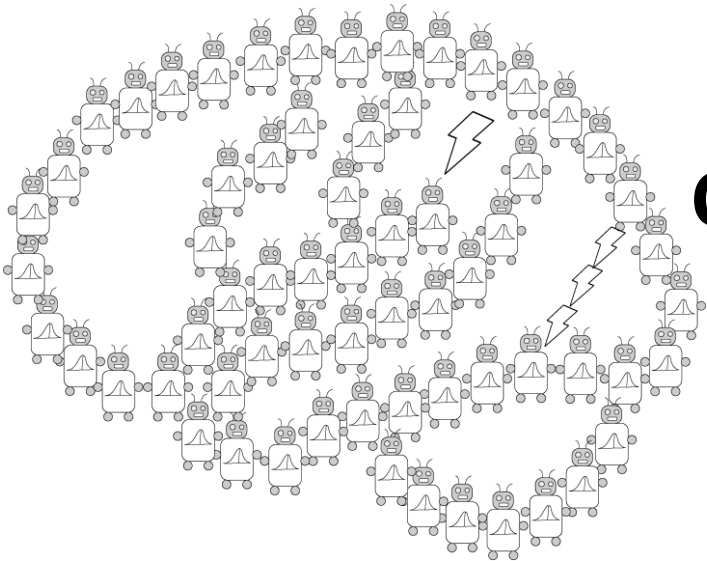


Bias-Variance Dilemma and an Introduction to Bayesian Nonparametrics



**PSYCH/COMP SCI/BMI 841:
Computational Cognitive Science**

Joseph Austerweil

April 10, 2025

- Admin Matters
- The bias-variance dilemma
- **Short rest 1**
- Intro to Bayesian Nonparametrics
- Gaussian Processes
- **Short rest 2**
- Feature construction
- One student presentation:
 1. Kemp & Tenenbaum 2003 by Zihan/Neuki

Administrative Matters

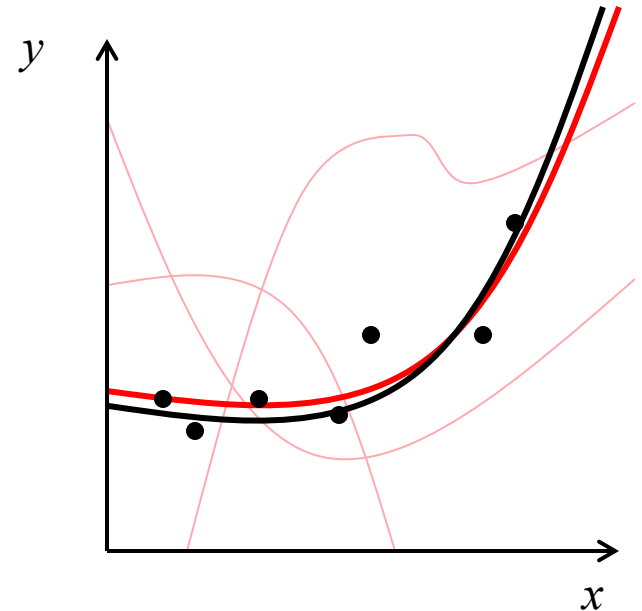
- Assignment 3 was due April 9 at 8:59pm
- Assignment 4 is due April 23 at 8:59pm
- 2 more discussion posts left
 - Only need to pass 5 through class for full credit
- Any unused quiz points will be free points to your final grade.
- Final project presentations May 1
 - Your project does NOT need to be complete by then
- Final project papers due May 6 at 8:59pm
- Questions?

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Learning mapping inputs to outputs

Regression:

- data D are n pairs (x, y)
- hypothesis space of functions, $y = g(x)$
- error: $C = (y - g(x))^2$
- learning algorithm picks function g using data D



Minimize prediction error

$$PE = E(y - g(x))^2 = \int (y - g(x))^2 p(x, y) dx dy$$

A learning strategy

Choose the function $g(x)$ in the hypothesis space that minimizes the training error

$$TE = \frac{1}{n} \sum_{i=1}^n (y_i - g(x_i))^2$$

Different hypothesis spaces define different learning algorithms

Neural networks correspond to the case where the hypothesis space is functions that can be produced by the network

Some hypothesis spaces

Linear functions

$$g(x) = p_1x + p_0$$

Quadratic functions

$$g(x) = p_2x^2 + p_1x + p_0$$

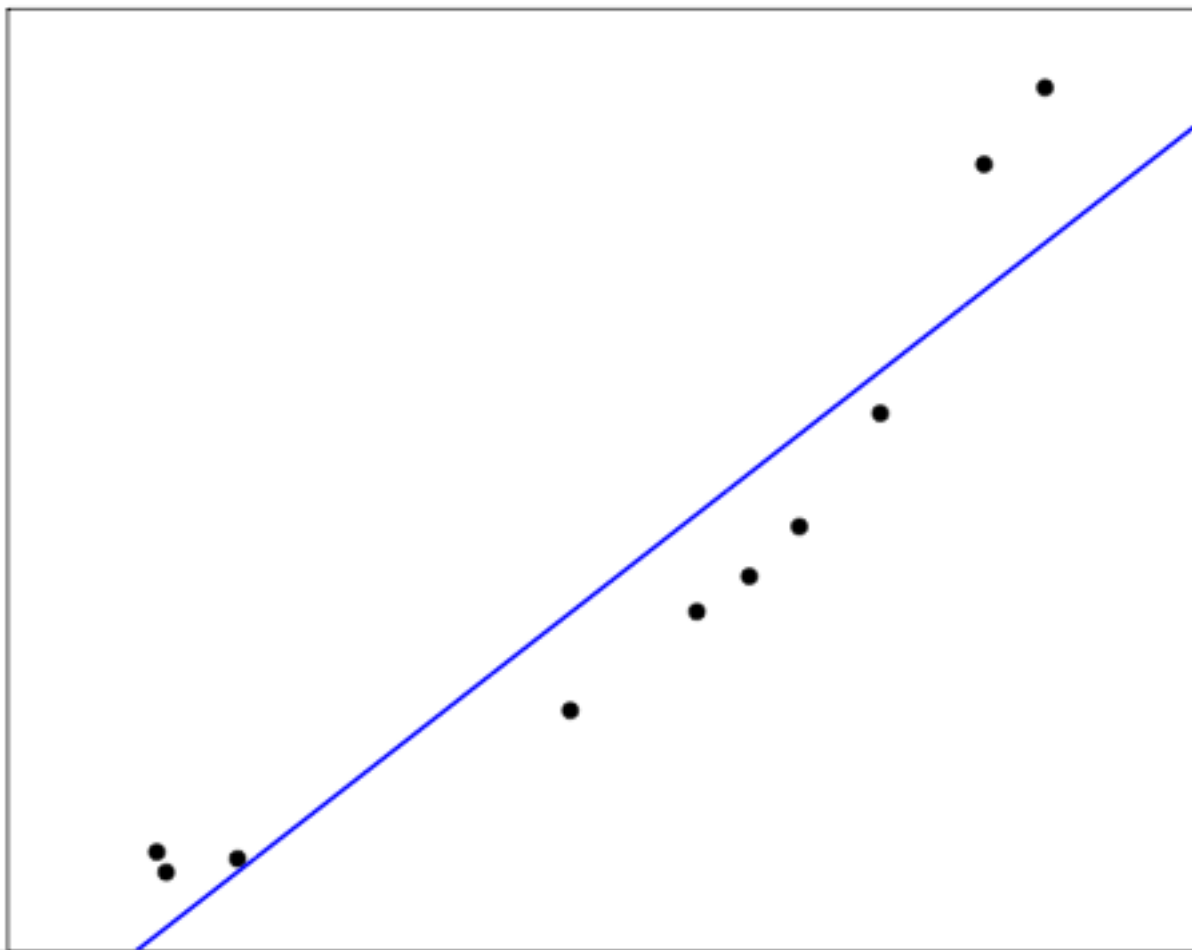
8th degree polynomials

$$g(x) = \sum_{j=0}^8 p_j x^j$$

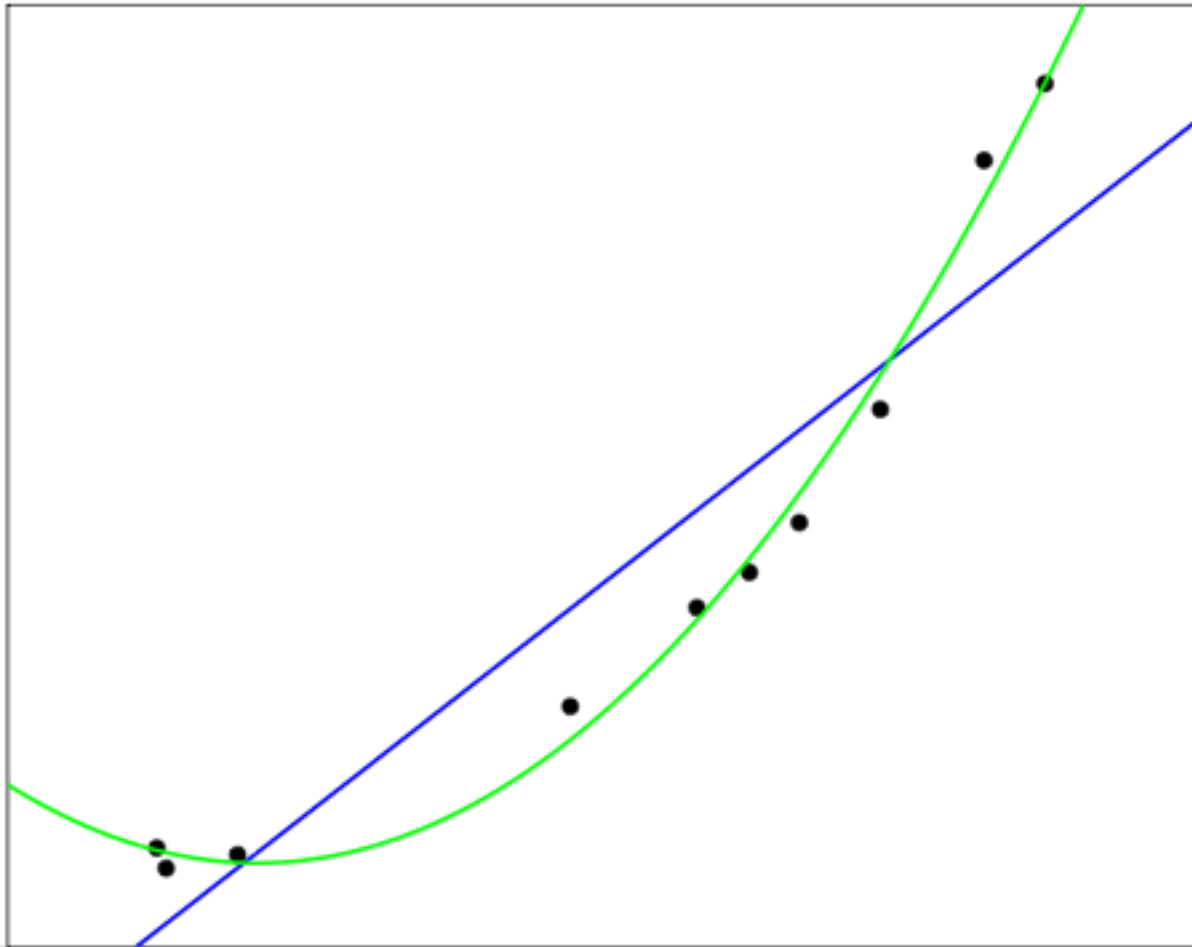
Minimizing training error



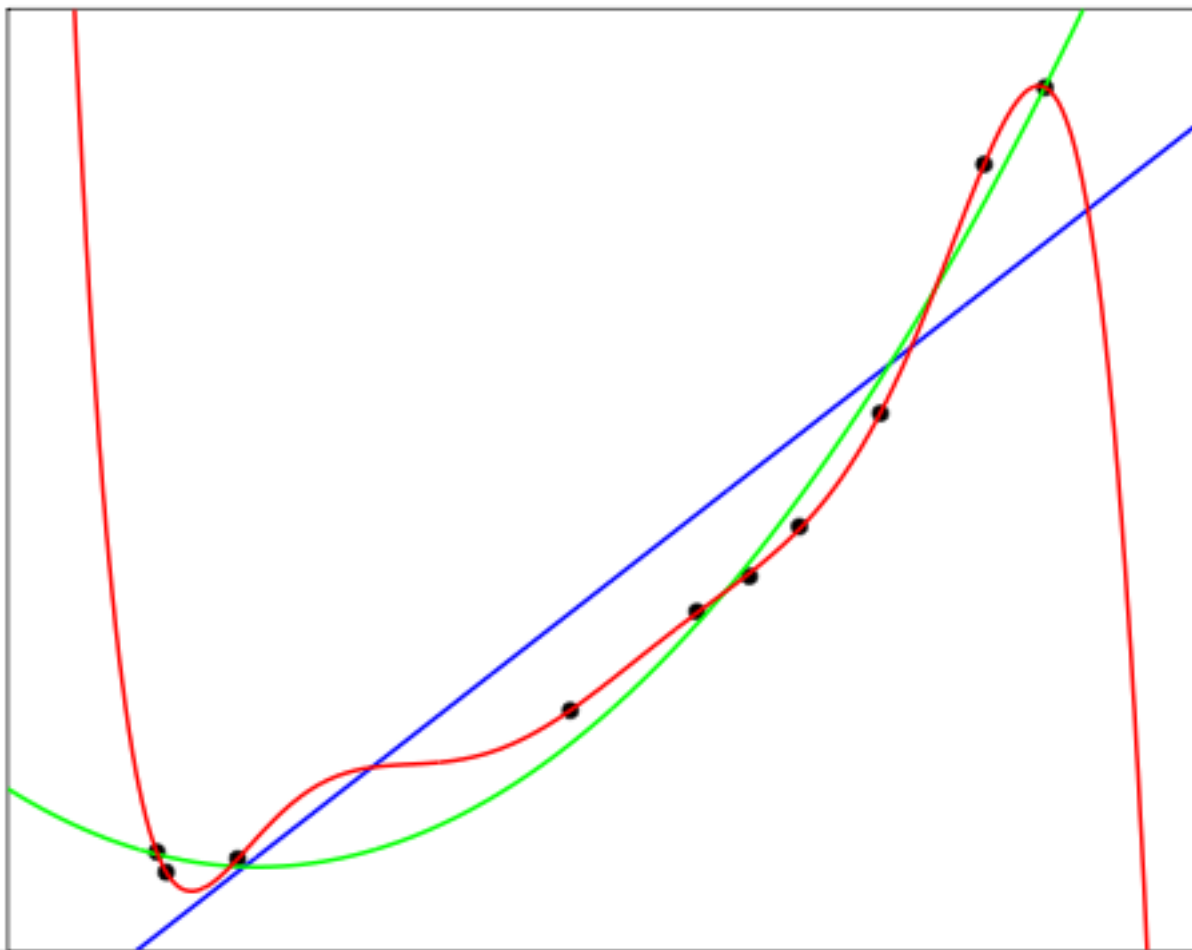
Minimizing training error



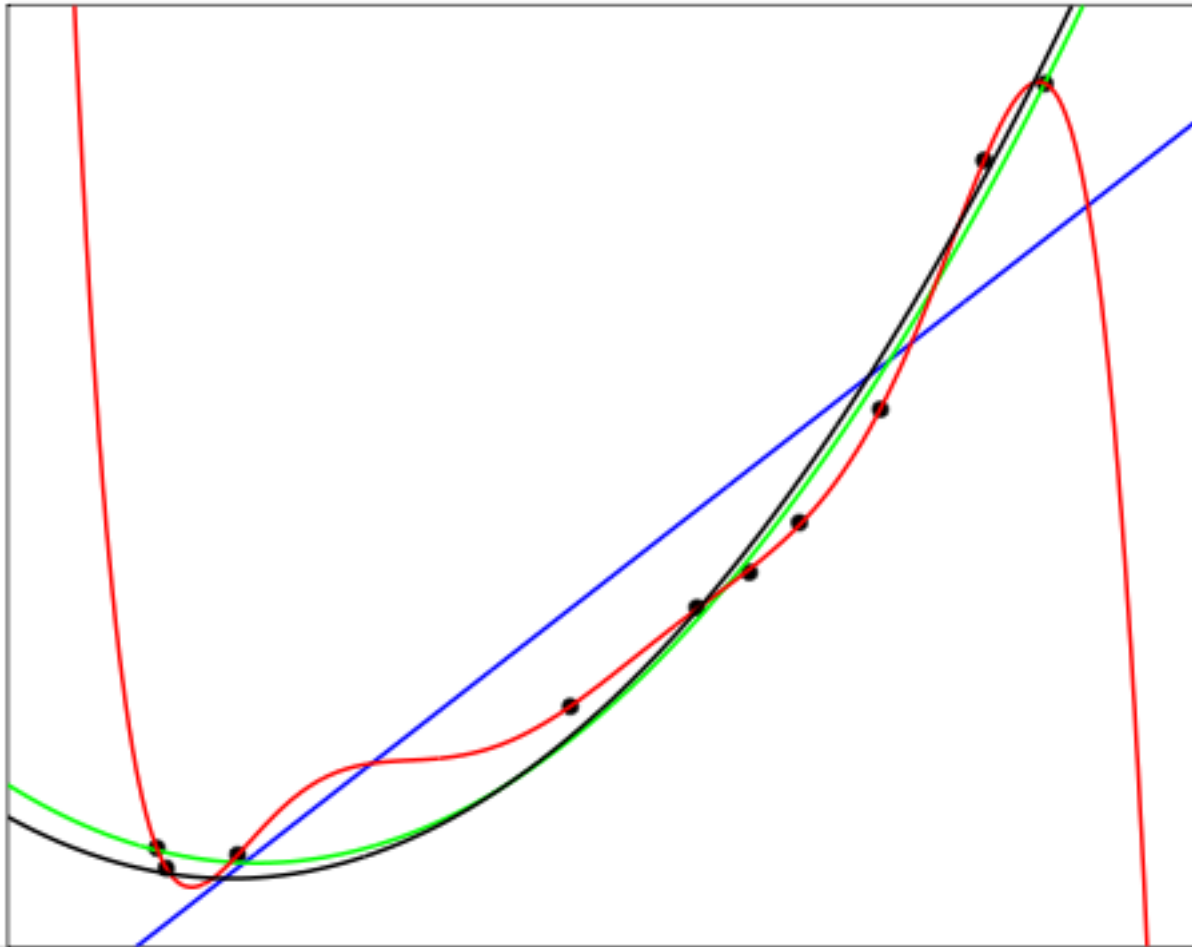
Minimizing training error



Minimizing training error



What about prediction error?



What happened?

The set of 8th degree polynomials contains almost all functions through 10 points.

Our data are some true function, plus noise.

Fitting the noise gives us the wrong function.

This is called *overfitting*.

More formally...

Assume $y = f(x) + \varepsilon$, where $E(\varepsilon) = 0$, $E(\varepsilon^2) = \sigma^2$
(assume error is normally distributed)

$$\begin{aligned} PE &= E_{p(x,y)}[(y - g(x))^2] \\ &= E_{p(x,y)}[(y - f(x) + f(x) - g(x))^2] \end{aligned}$$

$$\begin{aligned} &[(y - f(x)) + (f(x) - g(x))]^2 \\ (A + B)^2 &= A^2 + 2AB + B^2 \end{aligned}$$

More formally...

Assume $y = f(x) + \varepsilon$, where $E(\varepsilon) = 0$, $E(\varepsilon^2) = \sigma^2$
(i.e., error is normally distributed)

$$\begin{aligned} PE &= E_{p(x,y)}[(y - g(x))^2] \\ &= E_{p(x,y)}[(y - f(x) + f(x) - g(x))^2] \\ &= E_{p(x,y)}[(y - f(x))^2] + E_{p(x,y)}[(f(x) - g(x))^2] \\ &= \sigma^2 + E_{p(x)}[(f(x) - g(x))^2] \end{aligned}$$

A good algorithm makes $(f(x) - g(x))^2$ small

Bias and variance

A good algorithm makes $(f(x) - g(x))^2$ small.

g is chosen on the basis of the data D .

Evaluate algorithms by the expectation of $(f(x) - g(x))^2$ over data D generated from f .

$$E_{p(D)}[(f(x) - g(x))^2] = \underbrace{(f(x) - E_{p(D)}[g(x)])^2}_{\text{bias}^2} + \underbrace{E_{p(D)}[g(x) - E_{p(D)}[g(x)]]^2}_{\text{variance}}$$

Bias: how similar is the true function to the estimated function?

Variance: how much is the estimated function expected to vary across possible observations?

Making things more intuitive...

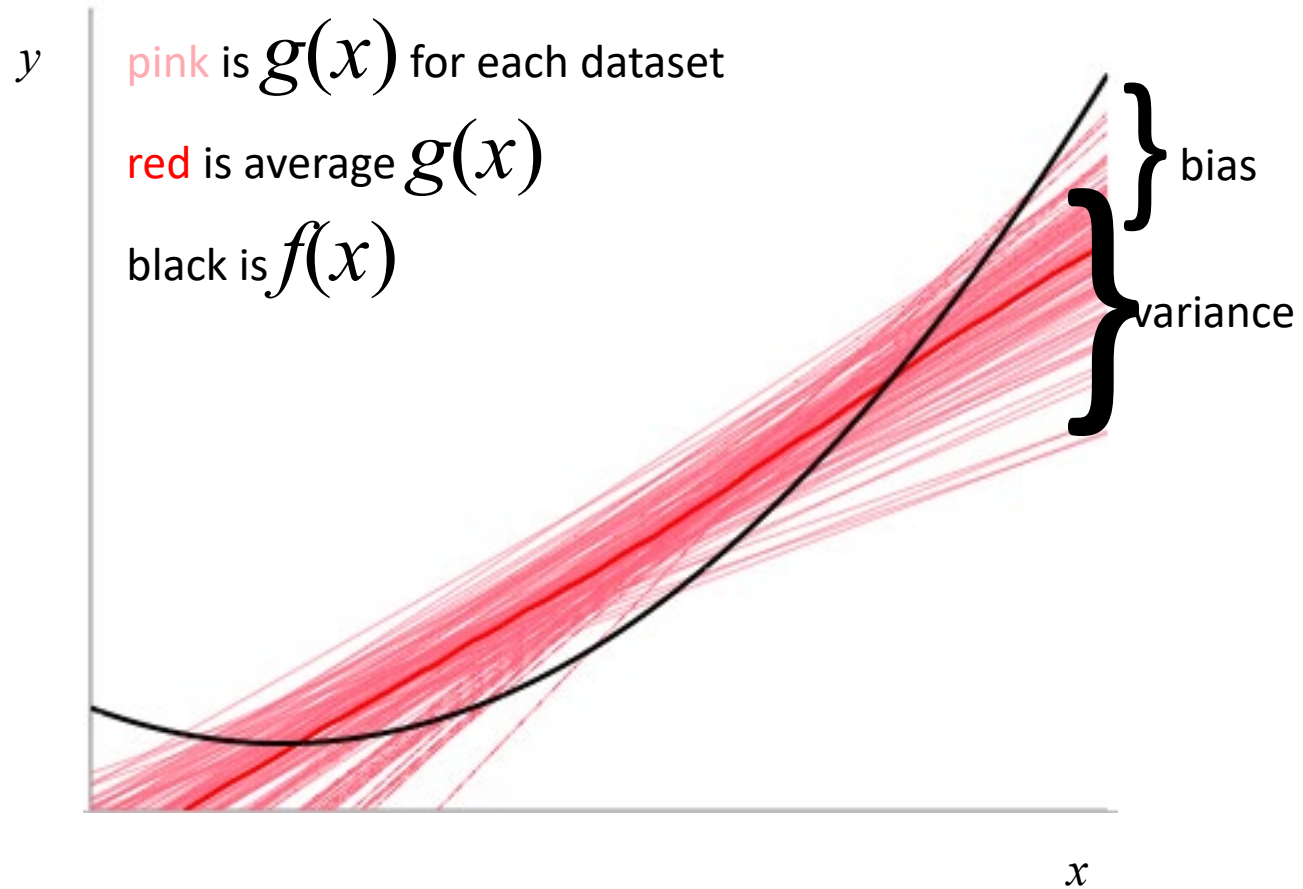
The next few slides were generated by:

- choosing a true function $f(x)$
- generating a number of datasets D from $p(x,y)$ defined by uniform $p(x)$, $p(y|x)f(x)$ plus noise
- finding the function $g(x)$ in the hypothesis space that minimized the training error

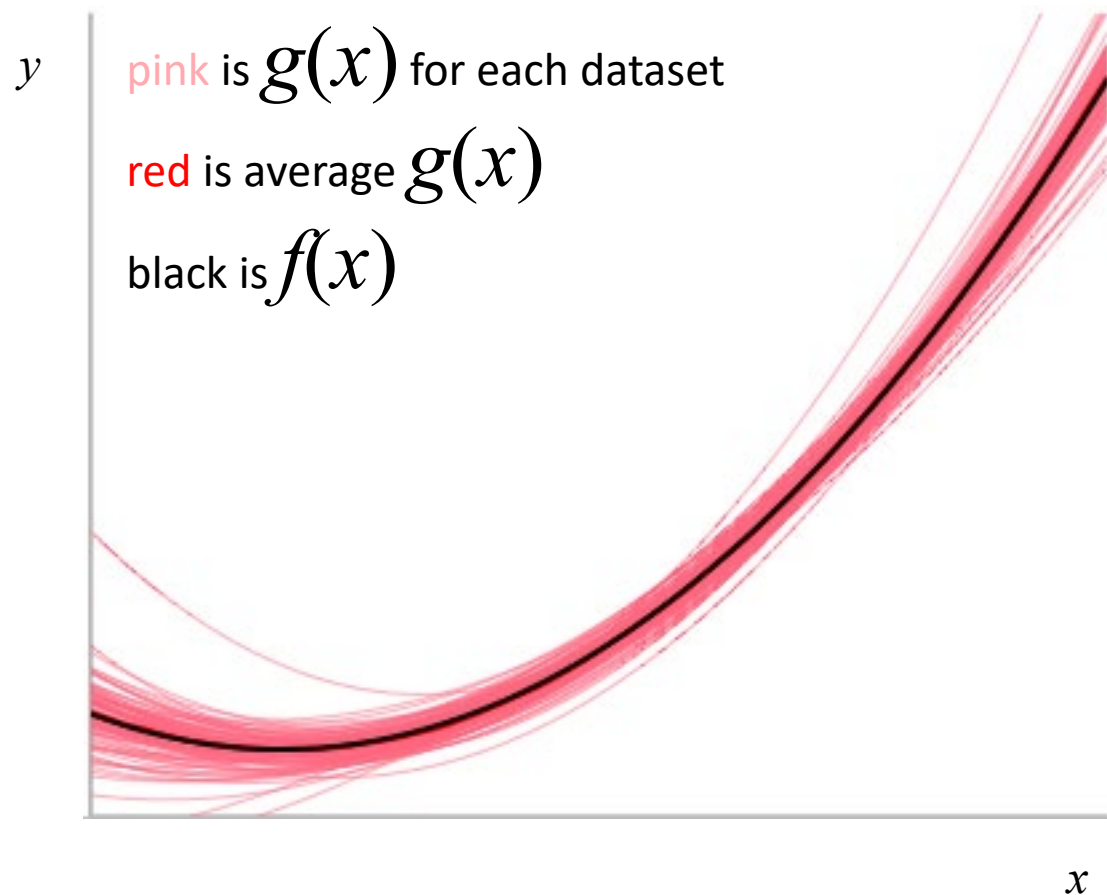
Comparing average of $g(x)$ to $f(x)$ reveals bias

Spread of $g(x)$ around average is the variance

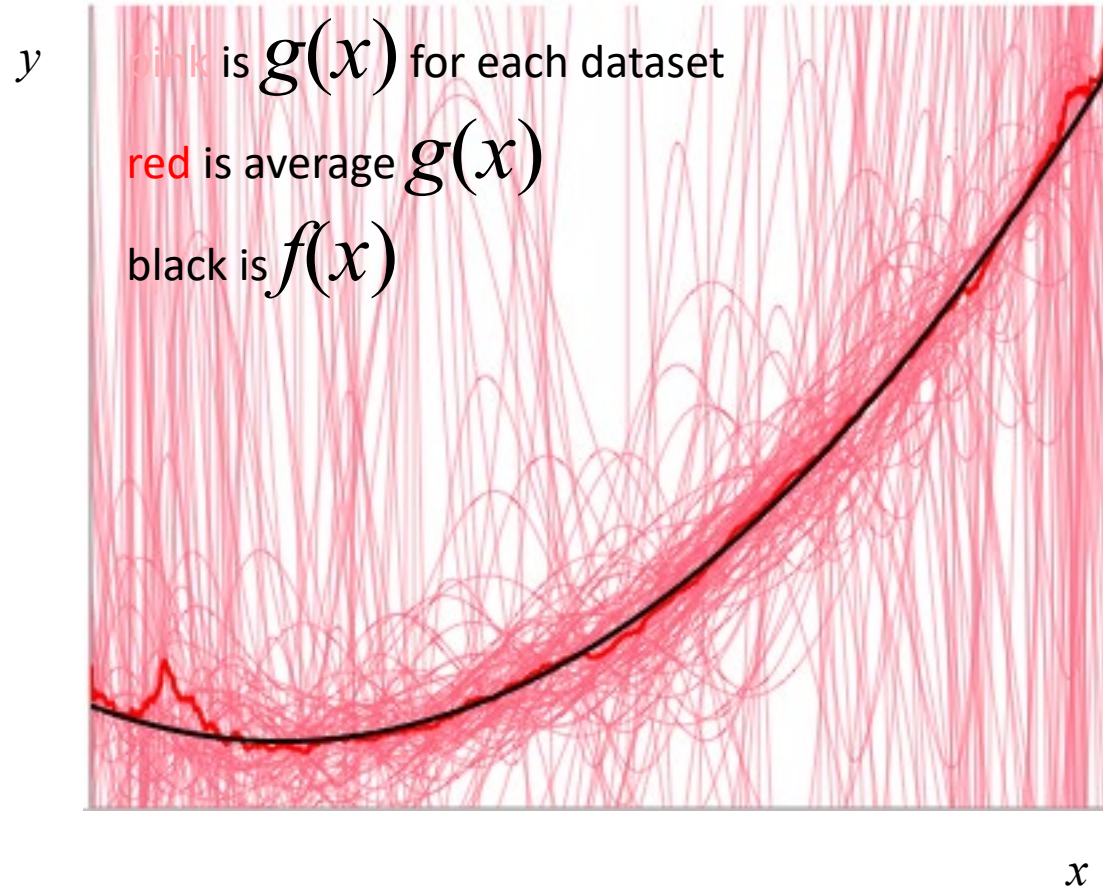
Linear functions ($n = 10$)



Quadratic functions ($n = 10$)



8-th degree polynomials ($n = 10$)



Bias and variance

(quadratic $f(x)$, with $n = 10$)

Linear functions

high bias, medium variance

Quadratic functions

low bias, low variance

8-th order polynomials

low bias, super-high variance

In general...

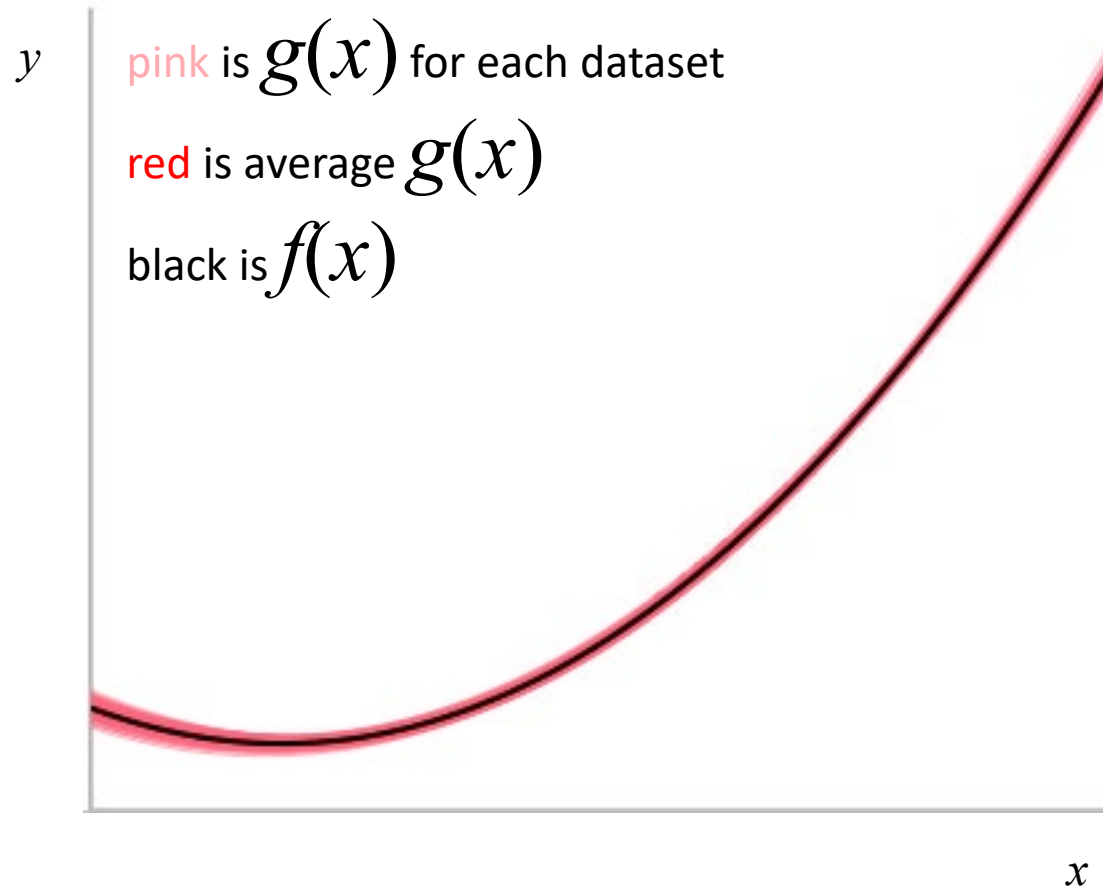
More flexible learning algorithms have higher variance, but low bias across several $f(x)$

The bias-variance dilemma or tradeoff:

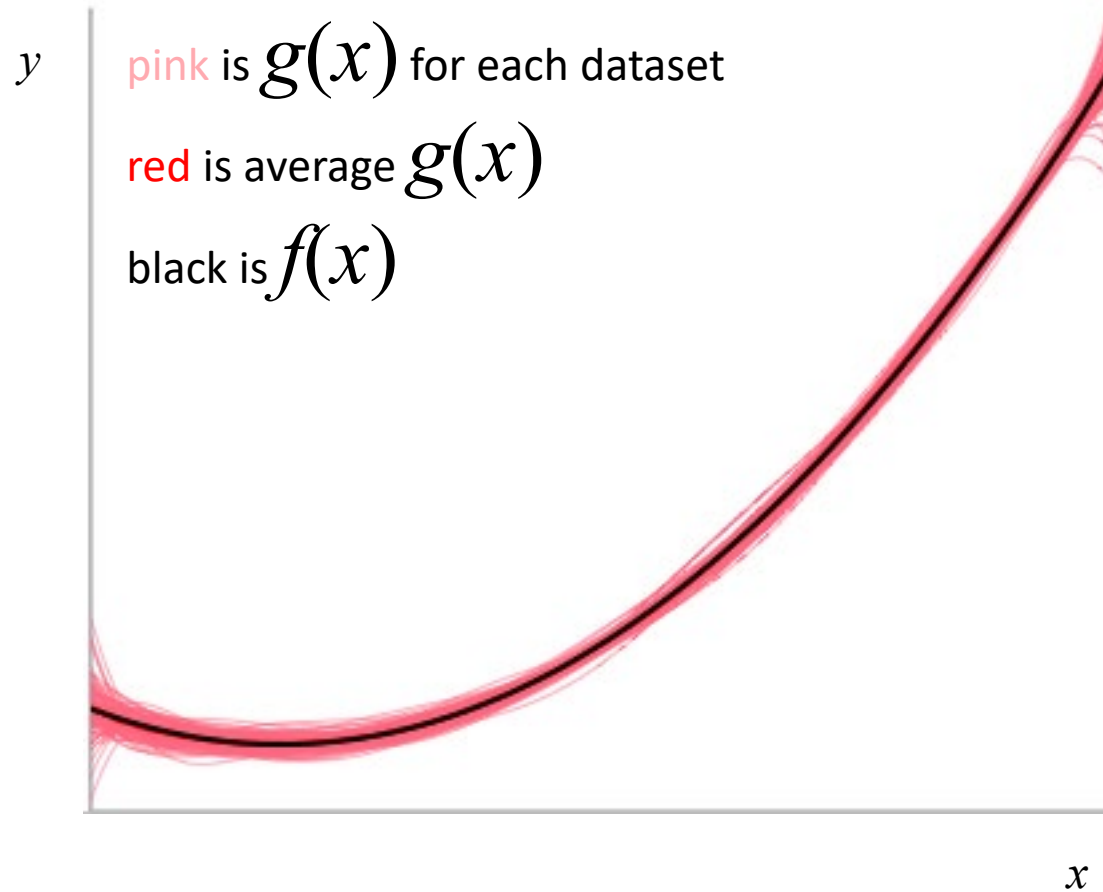
- if we want an algorithm that has low bias on a range of problems, we pay a price in variance

Note: This is mainly an issue when n is small

Quadratic functions ($n = 100$)



8-th degree polynomials ($n = 100$)



The moral

General purpose learning mechanisms do not work well with small amounts of data

So, the most flexible algorithm isn't always the best.

To make good predictions from small amounts of data, you need algorithms with bias that matches the problem being solved

What does this mean for learning agents?

The moral

General purpose learning mechanisms do not work well with small amounts of data.

So, the most flexible algorithm isn't always the best.

To make good predictions from small amounts of data, you need algorithms with bias that matches the problem being solved.

What does this mean for learning agents?

This suggests a different approach to studying induction might be profitable.

(what people learn as $n \rightarrow 0$, rather than $n \rightarrow \infty$)

Short rest 1

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What is a Bayesian Nonparametric Model?

Nonparametric - a model that can encode any* probability distribution (using arbitrarily complex representations).

E.g., Kernel methods

Bayesian - priors constrain which latent representations are preferred.

Bayes NP are nonparametric models that prefer to use simpler representations.

Types of latent structure and corresponding Bayes NP prior:

One discrete per datum (categories, trees, etc.) -> Dirichlet process

Multiple discrete per datum (features) -> Beta process

Continuous/Functions -> Gaussian process

Categorization as Mixture Modeling



To make an animal, first sample which animal we want from $\theta = (.2, .4, .4)$

Each animal is created from its associated ideal “snout size” μ_c and variance σ_c^2 .

This leads to the following *generative process* for each animal

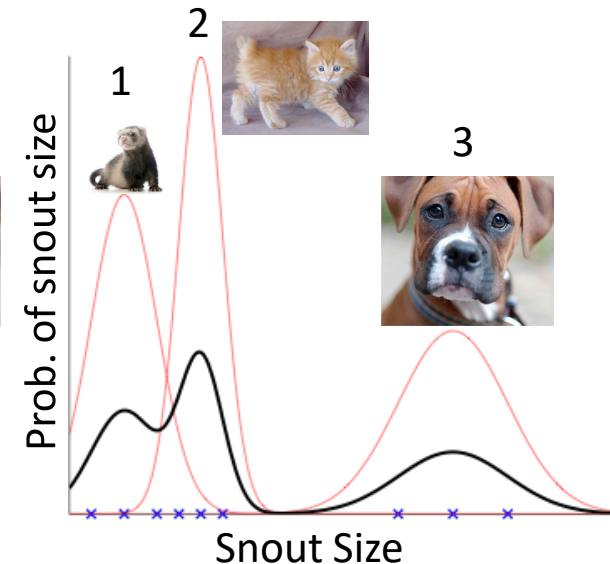
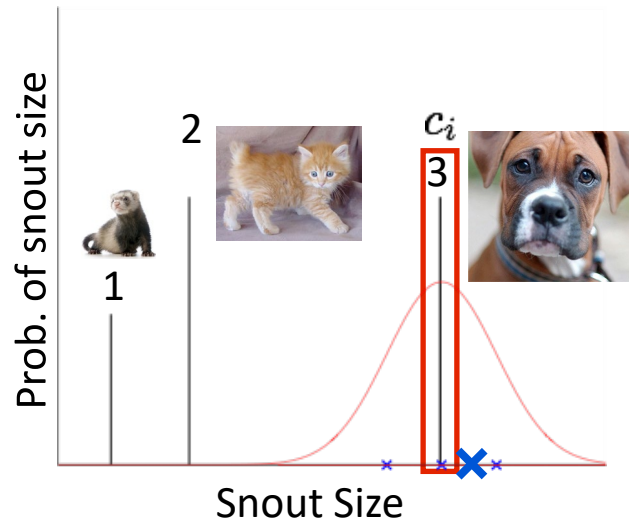
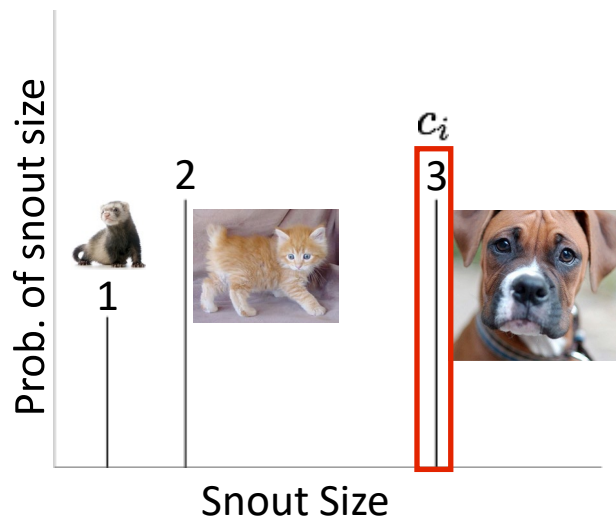
$$c_i | \theta \sim \text{Discrete}(\theta)$$

$$x_i | c_i \sim \text{Gaussian}(\mu_{c_i}, \sigma_{c_i}^2)$$

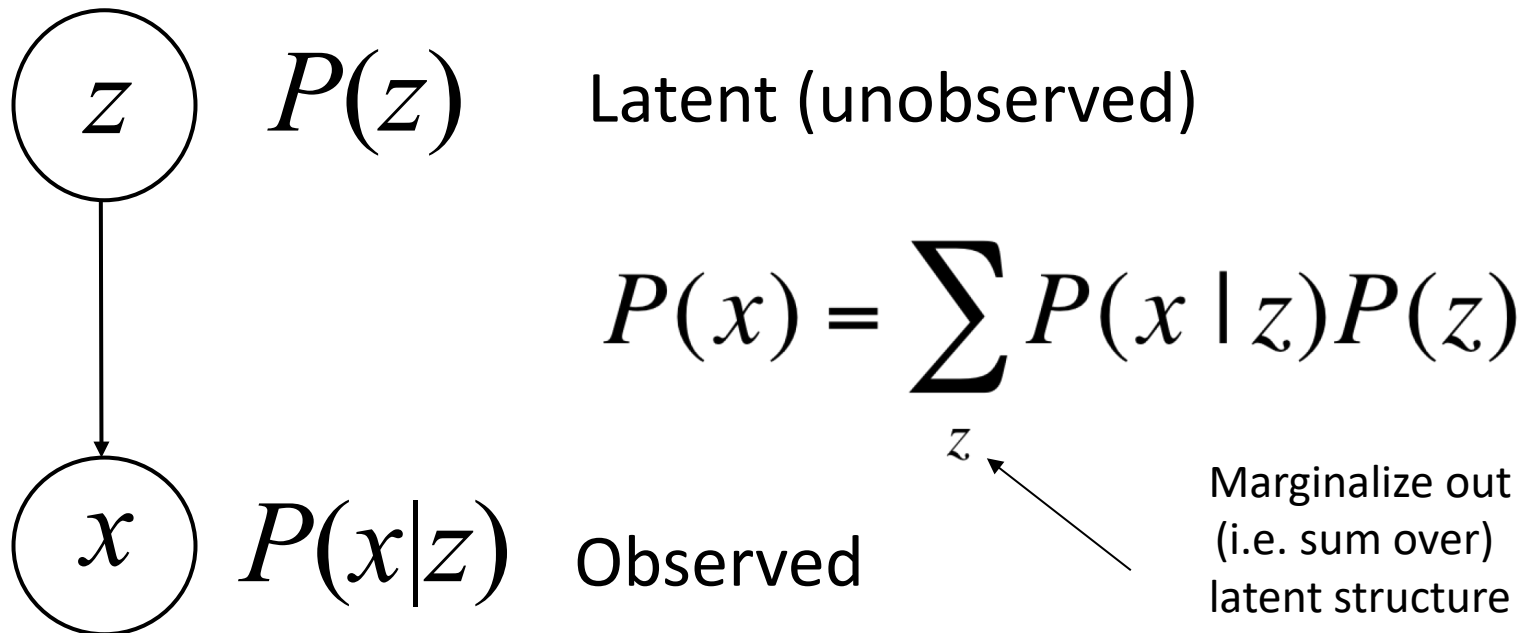
Prob. of new snout size:
$$P(x_i) = \sum_{k=1}^3 P(x_i, c_i = k) = \sum_{k=1}^3 P(x_i | \mu_k, \sigma_k^2) P(c_i = k | \theta)$$

Categorize a new animal given its snout size:

$$P(c_i = 1 | x_i, \theta) = \frac{p(x_i | \mu_1, \sigma_1^2) P(c = 1 | \theta)}{\sum_{k=1}^3 p(x_i | \mu_k, \sigma_k^2) P(c = k | \theta)}$$



Latent variable models



z is each x 's cluster index

How do we put a prob. on clusterings?

But how do we pick the number of clusters?

Probability distributions over clusterings!

A **clustering** (partition) splits data into K mutually exclusive blocks.

The **Chinese Restaurant Process** (α)

Restaurant with infinite number of infinite capacity tables.

Generative process via culinary metaphor:

1. First customer starts their own table.
2. Customer n sits at a table proportional to the number of customer sitting there, or creates a new table with probability $(\alpha / (\alpha + n - 1))$

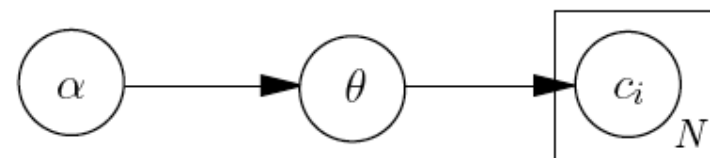
A (in)finite mixture model

Suppose there are N objects and K clusters.

Let $c_{nk} = 1$ denote that object n is in cluster k .
(each object is in one cluster)

Let θ_k define the probability an object is in cluster k .

Let m_k be the number of objects in cluster k .



$$\vec{\theta}|\alpha \sim \text{Dirichlet}(\alpha_1/K, \dots, \alpha_K/K)$$

$$\vec{c}_n|\vec{\theta} \sim \text{Multinomial}(\vec{\theta})$$

$$P(\mathbf{C}|\alpha) = \int d\vec{\theta} P(\vec{\theta}|\alpha) \prod_{n=1}^N P(\vec{c}_n|\vec{\theta}) = \frac{\Gamma(\alpha)}{\Gamma(N + \alpha)} \frac{\prod_{k=1}^K \Gamma(m_k + \alpha/K)}{\Gamma(\alpha/K)^K}$$

$$\lim_{K \rightarrow \infty} P(\mathbf{C}|\alpha) \stackrel{*}{=} \frac{\alpha^{K_+} \Gamma(\alpha)}{\Gamma(N + \alpha)} \prod_{k=1}^{K_+} (m_k - 1)! \quad \text{where } K_+ \text{ is the number of "used" clusters.}$$

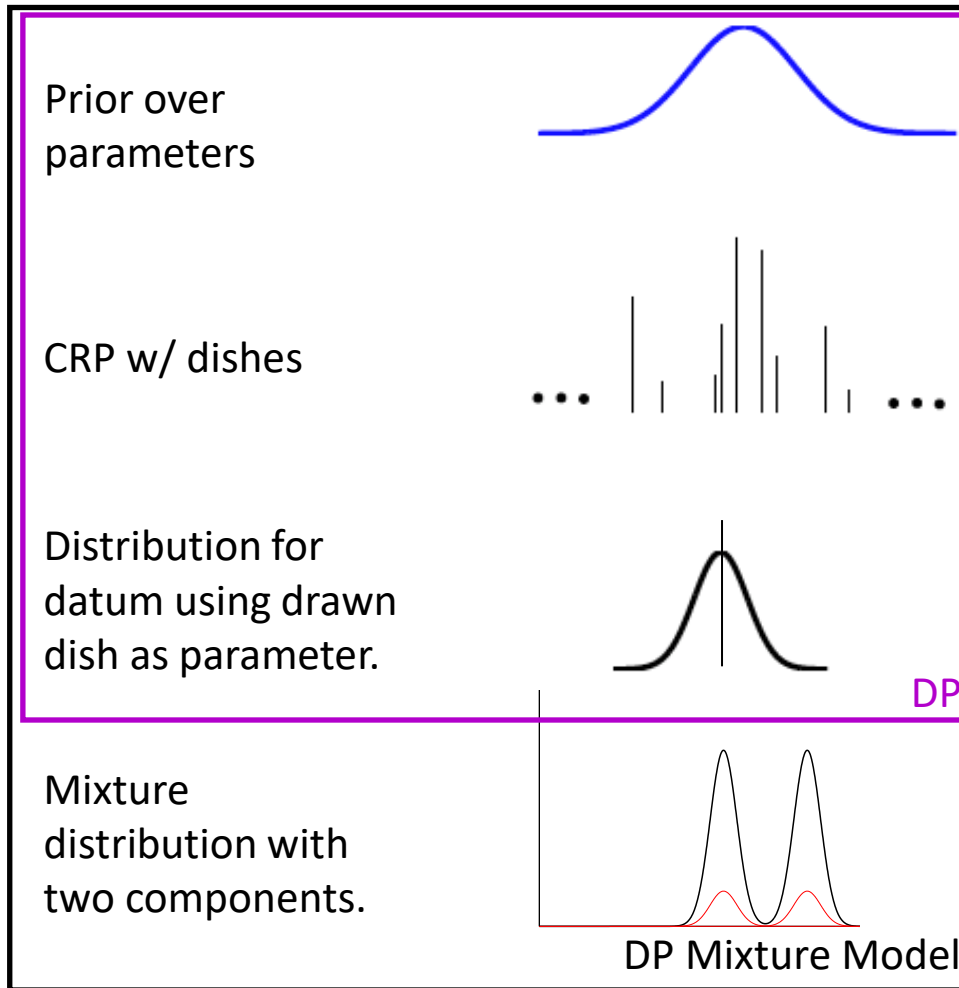
This probability distribution is equivalent to the one given by the Chinese restaurant process (CRP).

Customers (objects) arrive at a restaurant with an infinite number of infinitely-sized tables (clusters).

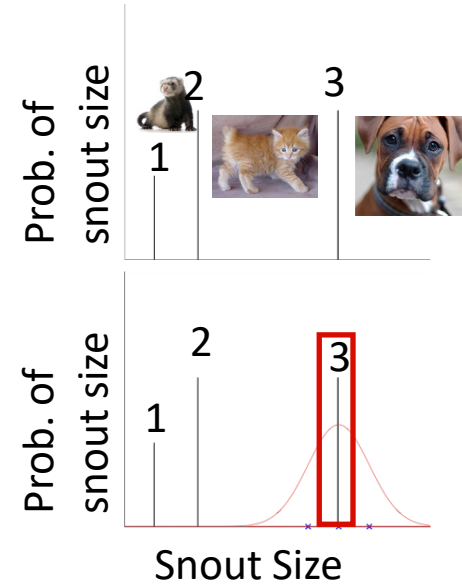
A customer sits down at an occupied table with probability proportional to the number of occupants.

A customer starts a new table with probability proportional to α .

Connecting the CRP to mixture models



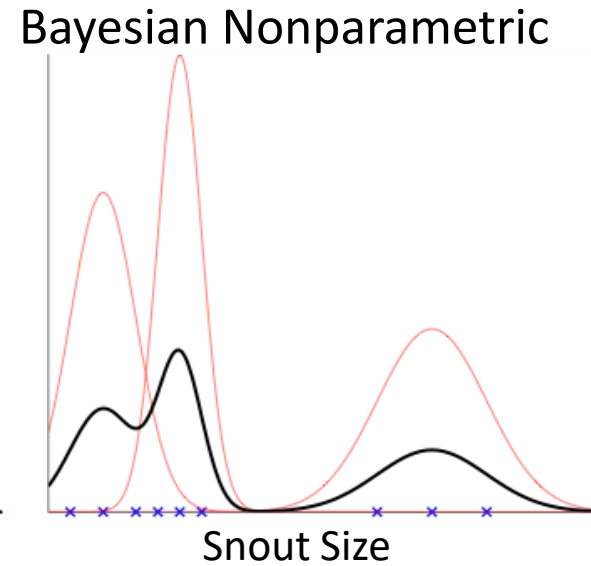
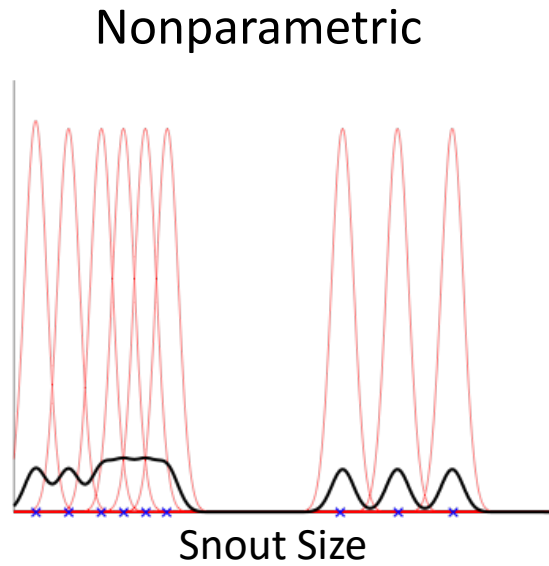
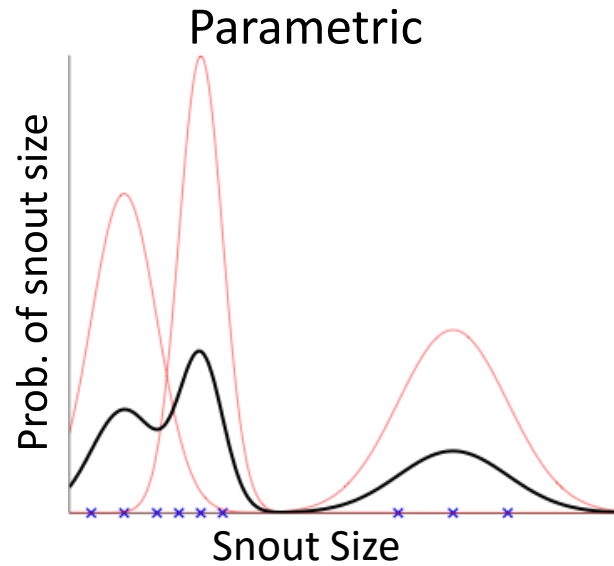
Gaussian prior on params.



When the tables are labeled with parameters drawn *iid* from some distribution, this generates the *Dirichlet process* or the *Polya Urn*.

(Jordan, 2010)

Categorization as Mixture Modeling



Definition: Restricted family of distributions

Can express *any* distribution

Structure grows with data.*

Examples: Finite Gaussian mixture models (prototypes)

Kernel methods (exemplar models)

Dirichlet process (rational categorization model)

Pros: Efficient learning;
More robust to noise

Can learn any distribution.

Can learn any distribution.
Less prone to overfitting.

Cons: Cannot represent every distribution.

Inefficient learning;
Prone to overfitting

Inference is hard.

What does nonparametric mean?
How does it differ from parametric?

Nonparametric models can learn any possible distribution.*

Parametric models are restricted to a family (e.g., Gaussians with different means).

What is “Bayesian” about Bayesian nonparametrics?

Bayesian nonparametric models can learn any possible distribution,
but they are *biased* towards particular *types* of distributions.

Are there no parameters in Bayesian nonparametric models?

Nonparametric models have parameters. They just have an infinite number of them!

Are Gibbs samplers a type of Bayesian nonparametric model?

Gibbs sampling is an inference method. It is applied to models.

What is the distinction between models and inference?

Models specify a probability distribution over data and possibly latent factors.
Inference methods sample from (or find the max) the posterior distribution.

*Nonparametric can also mean models whose structure grow with the amount of data.
These two definitions are strongly, but not perfectly correlated.

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One-Dimensional GP

Lets look at a simple* one-d GP:

$$f(x) \sim GP \left(m(x) = 0, k(x, x') = \exp \left\{ -\frac{1}{2} (x - x')^2 \right\} \right)$$

Well what's that mean?

Lets say we want the value of f at $x_1, \dots, x_N$.

Well we actually have enough info to sample...

$$(x_1, \dots, x_N) = (-5, -4.8, \dots, 0, \dots 4.8, 5)$$

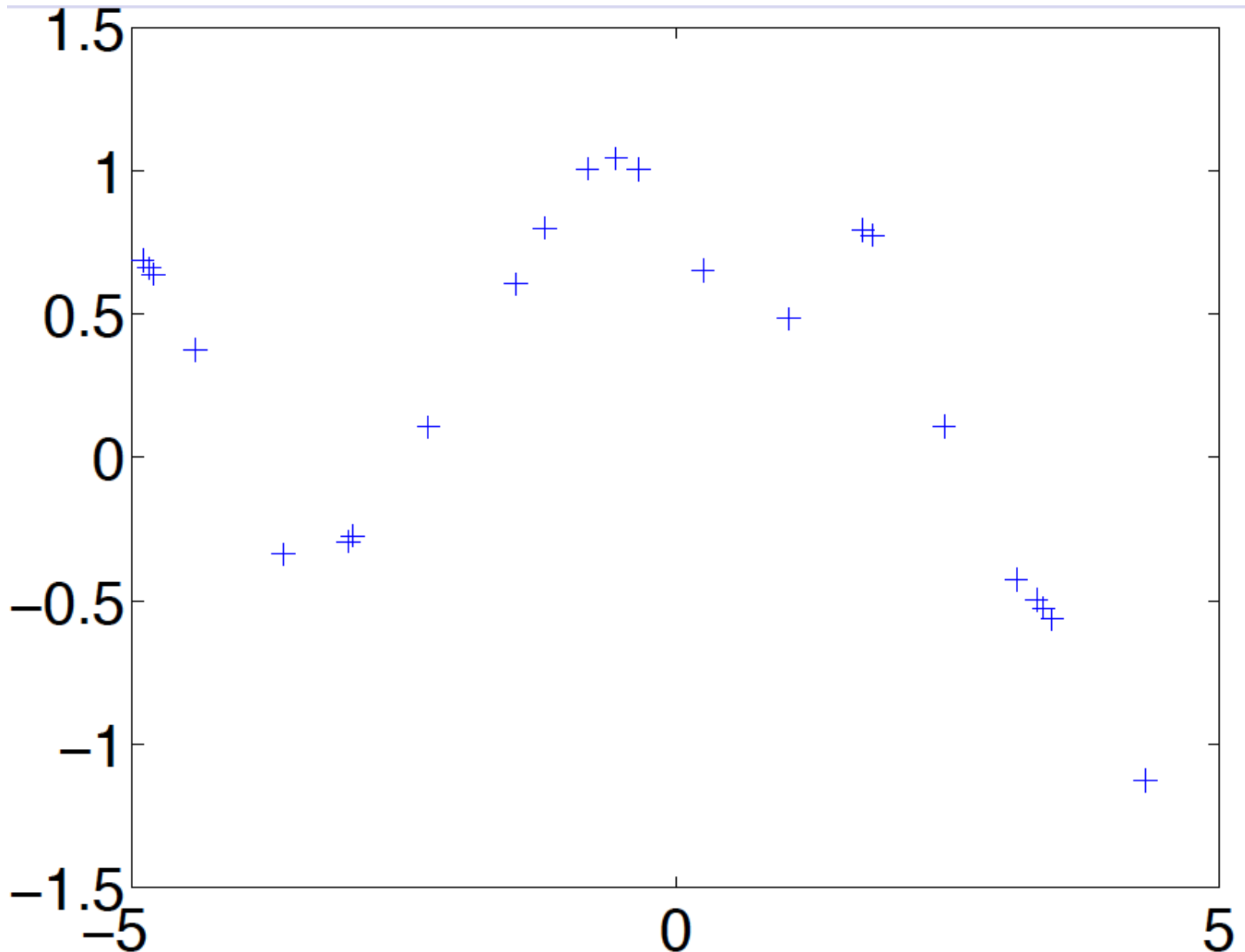
$$(f(x_1), \dots, f(x_N)) \sim N(0, \Sigma)$$

$$\Sigma_{ij} = k(x_i, x_j)$$

*simple isn't even **that** simple

From Rasmussen's 2006 NIPS Tutorial

One-Dimensional GP



From Rasmussen's 2006 NIPS Tutorial

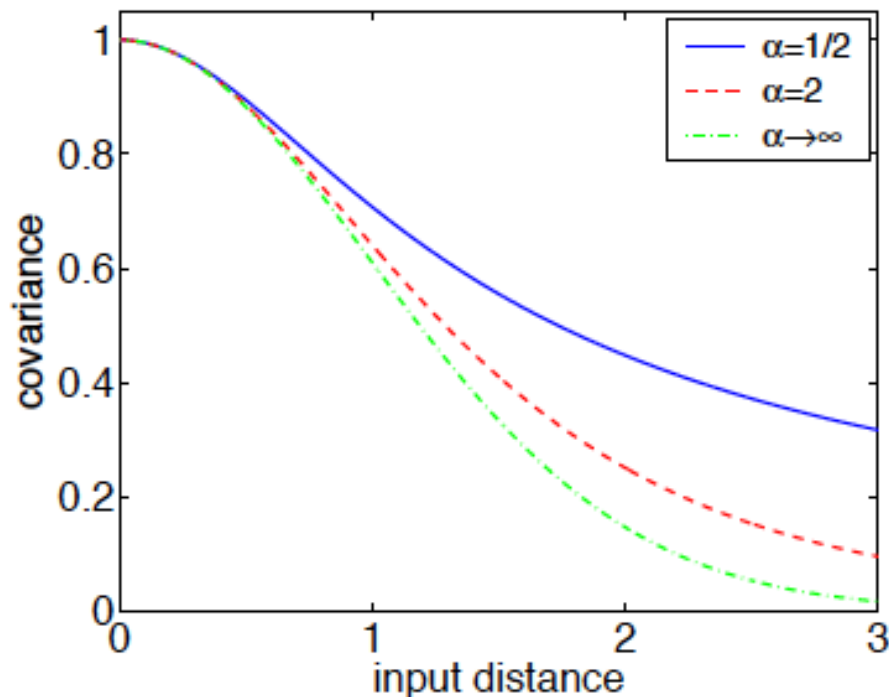
Lots of different kernel functions

We already met the Gaussian covariance function.

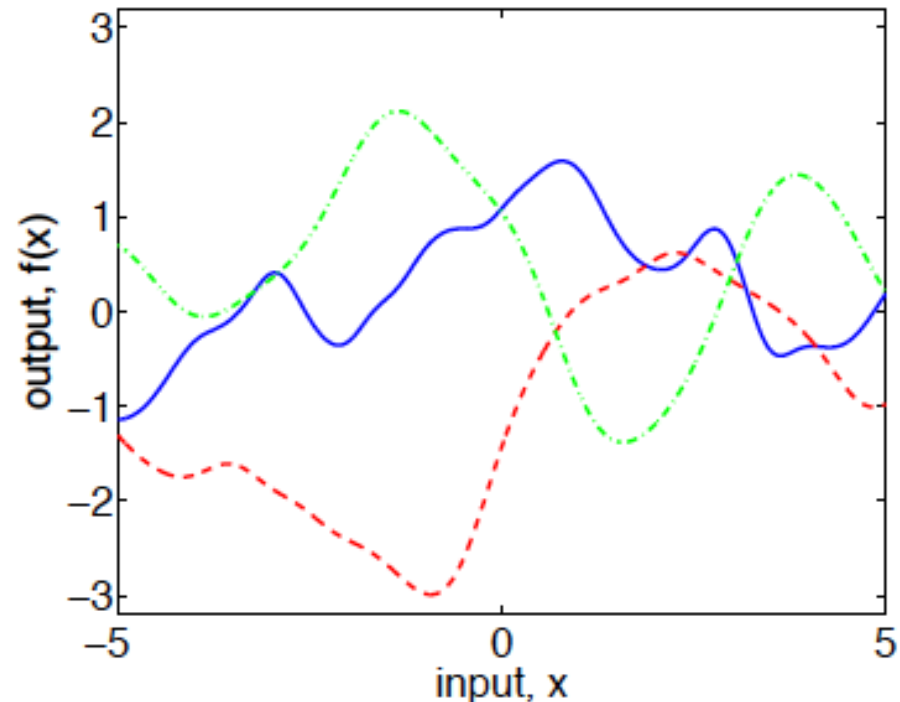
Another is the *Rational Quadratic*

$$k_{\text{RQ}}(x_i, x_j) = \left(1 + \frac{(x_i - x_j)^2}{2\alpha l^2} \right)^{-\alpha}$$

Kernel behavior



Sampled functions



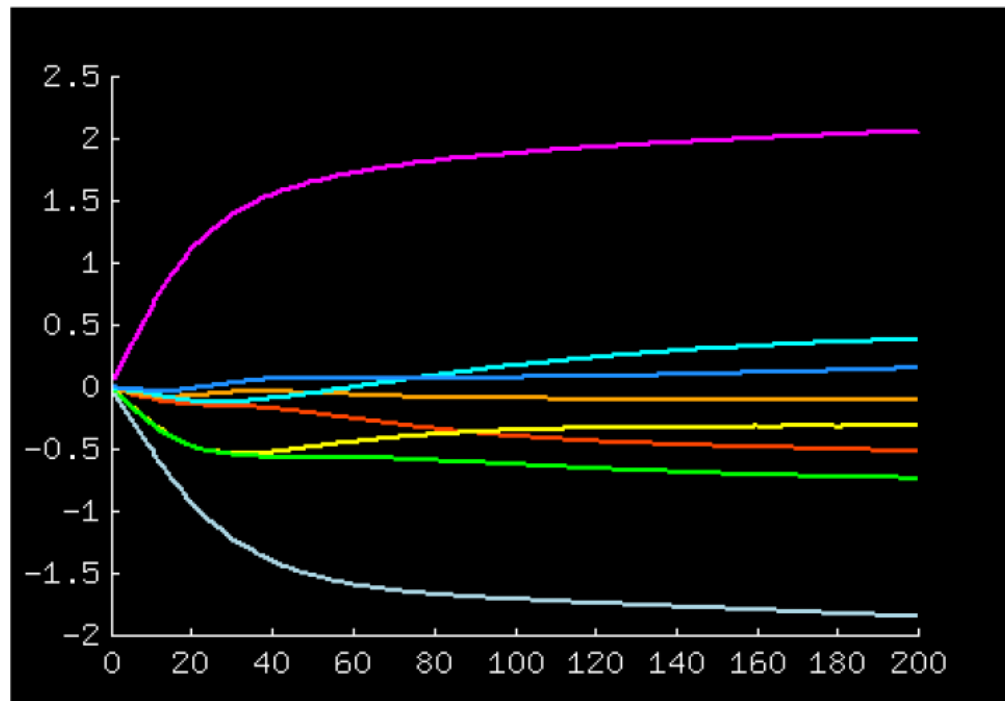
From Rasmussen's 2006 NIPS Tutorial

What about Neural Networks (NN)?

A NN with 1 infinite hidden layer is a GP with kernel function.

$$k_{NN}(\mathbf{x}_i, \mathbf{x}_j) = \frac{2}{\pi} \sin^{-1} \left(\frac{2\mathbf{x}_i^T \Sigma \mathbf{x}_j}{\sqrt{(1 + 2\mathbf{x}_i^T \Sigma \mathbf{x}_i)(1 + 2\mathbf{x}_j^T \Sigma \mathbf{x}_j)}} \right)$$

Sampled functions



From Mackay's Tutorial

- <https://chi-feng.github.io/gp-demo/>

Short rest 2

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Representing input is hard



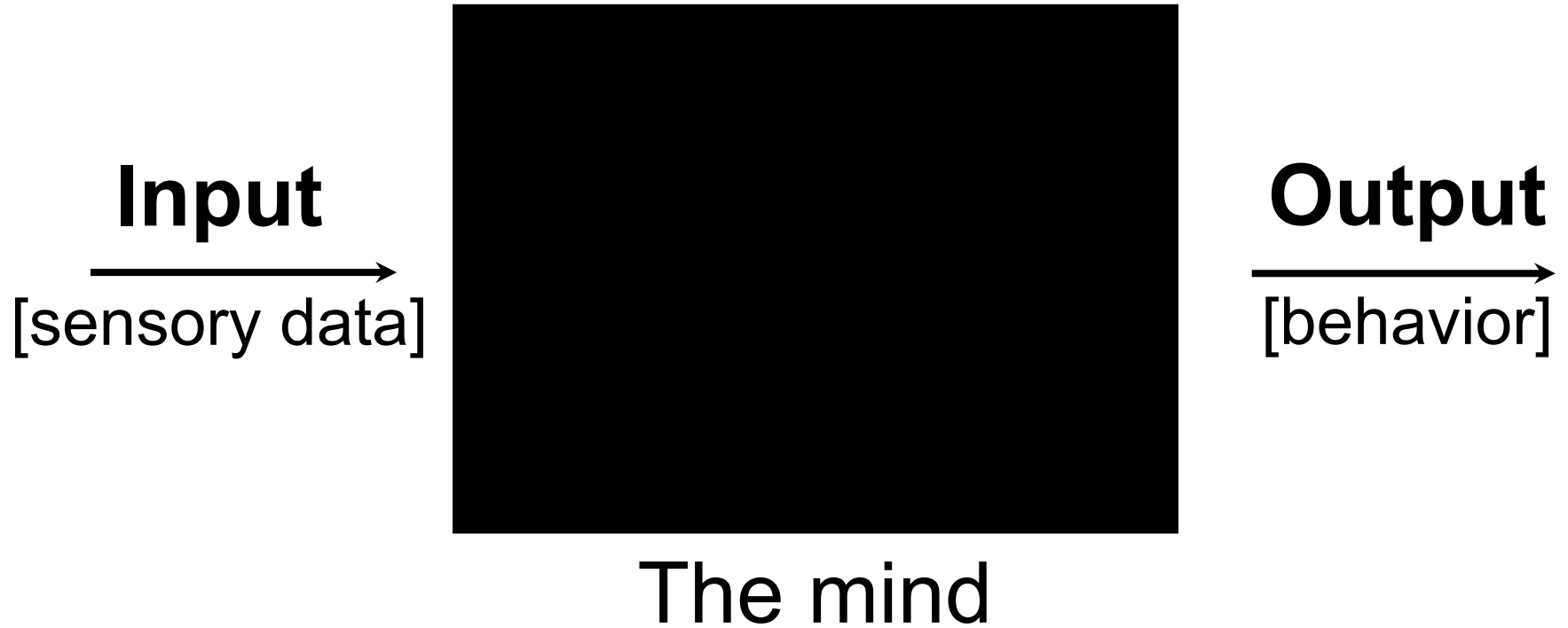
Rubin, Nakayama, and Shapley (2002)

Representing input is hard

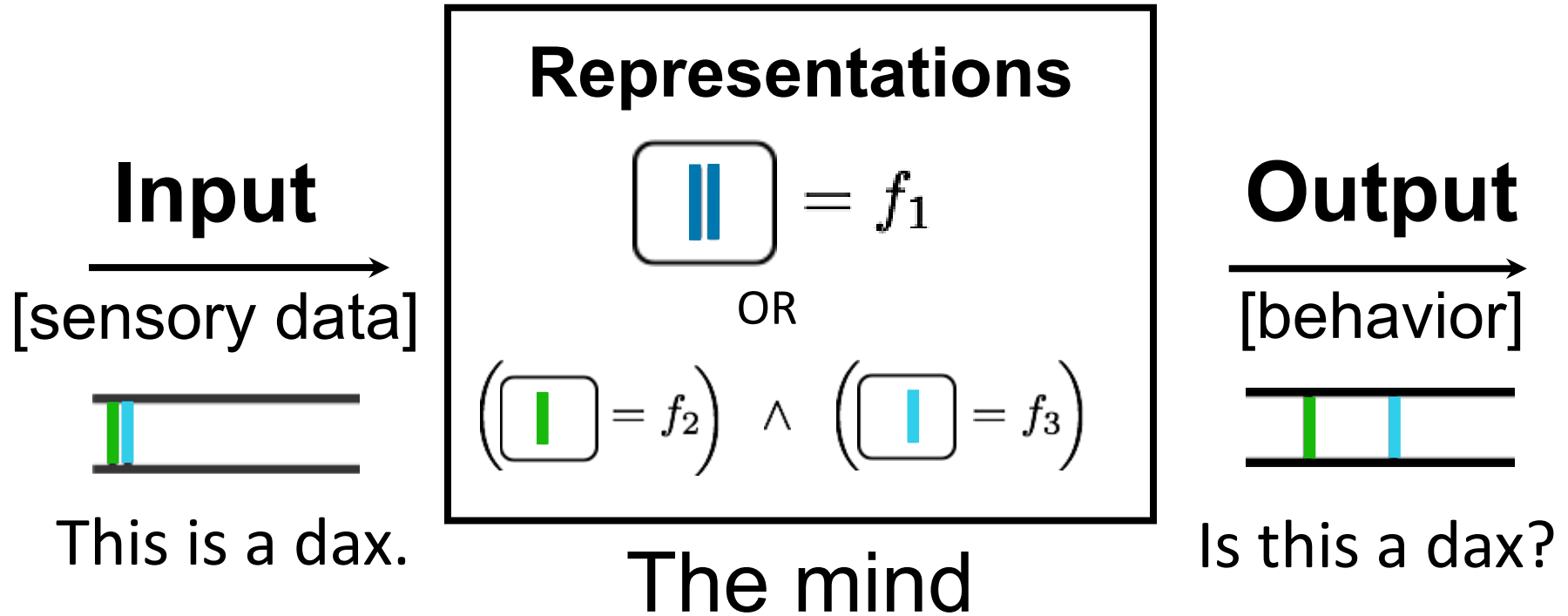


Rubin, Nakayama, and Shapley (2002)

Computational cognitive psychology



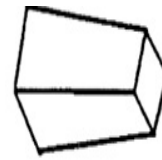
Computational cognitive psychology



How do people infer useful features?

Six criteria for a computational model of human feature inference

1. Primitives from sensory input (Hoffman & Richards, 1985; Schyns et al., 1998)



geon



Frank Gehry

2. Unlimited features (Goldmeier, 1936; Goodman, 1972; Murphy & Medin, 1985)

*A mug has a handle,
is heavier than a mosquito,
is smaller than a bread box...*

How do people infer useful features?

Six criteria for a computational model of human feature inference

3. Context sensitivity

(Garner, 1974; Palmer, 1975;
Selfridge, 1955; Tversky, 1977)

TAE CAT

4. Prior expectations

(Chater & Vitanyi, 2003; Goldstone, 2000;
Hochberg & McAlister, 1953)



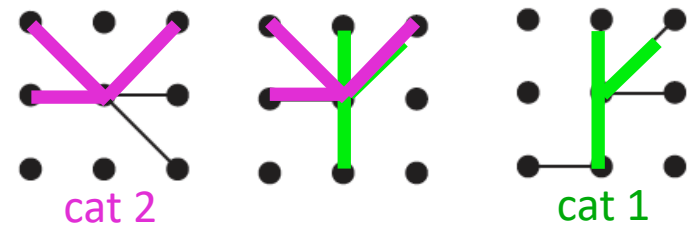
(Austerweil & Griffiths, 2013)

How do people infer useful features?

Six criteria for a computational model of human feature inference

5. Category diagnosticity

(Petzow & Goldstone, 1994; Schyns & Murphy, 1994)



6. Transformational invariance

(Palmer, 1983; Rock, 1973; Rust & Stocker, 2010)

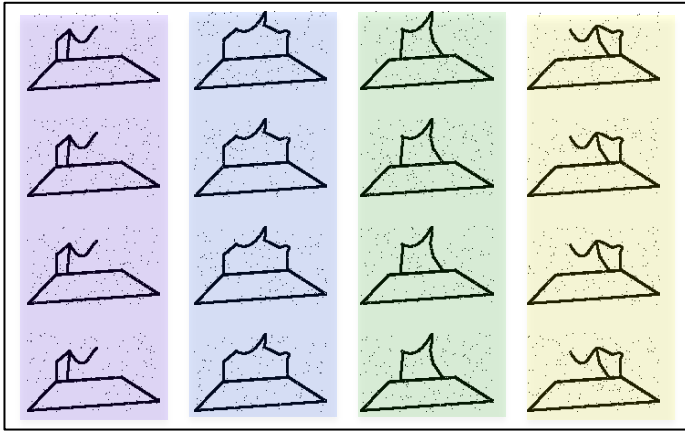


Substitutive, incremental learning, hierarchies...

(Austerweil & Griffiths, 2013)

Feature inference

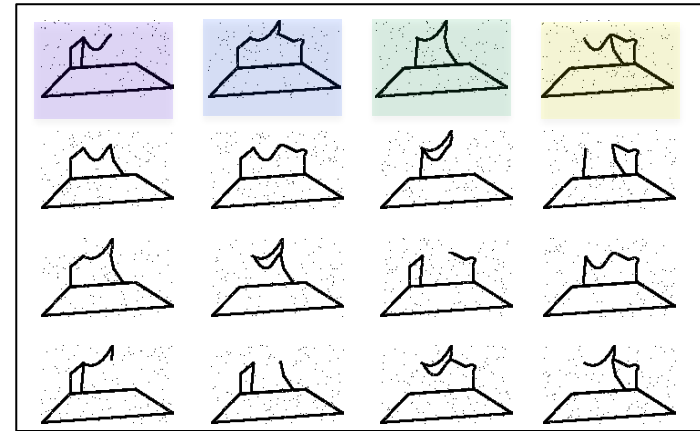
Inscriptions found on a Martian cave wall:



Same cave?



Same cave?



Same cave?

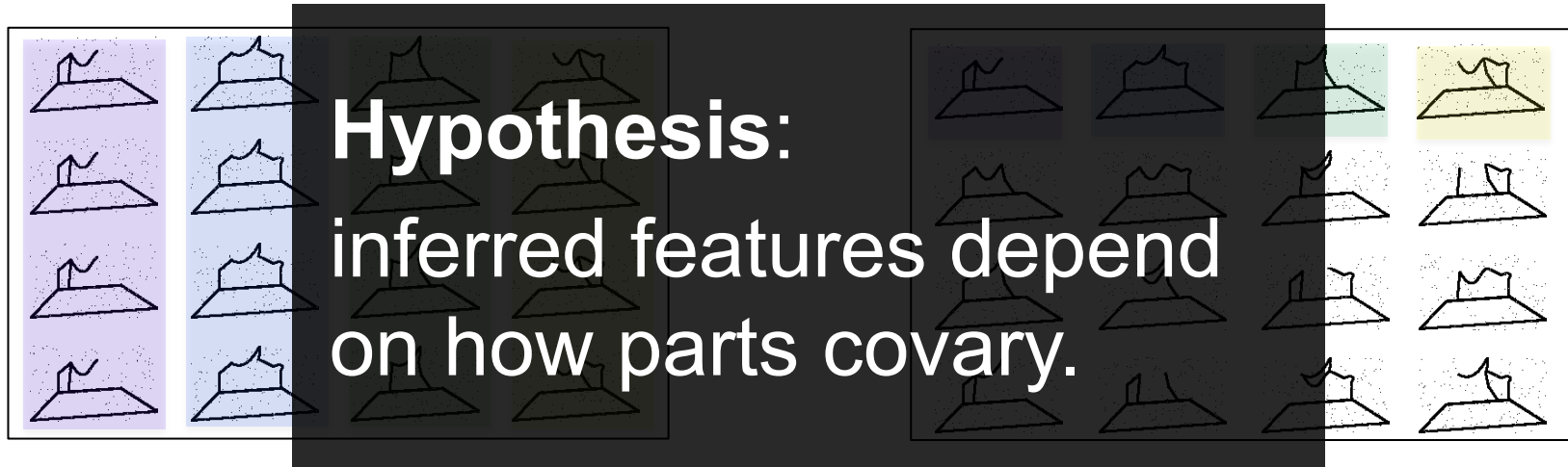


Same cave?



Feature inference

Inscriptions found on a Martian cave wall:



Correlated parts



represented as a
single feature

Independent parts



represented as
separate features

Generating experiment stimuli

Each image is composed of 1 shared part and 3 out of 6 other parts.

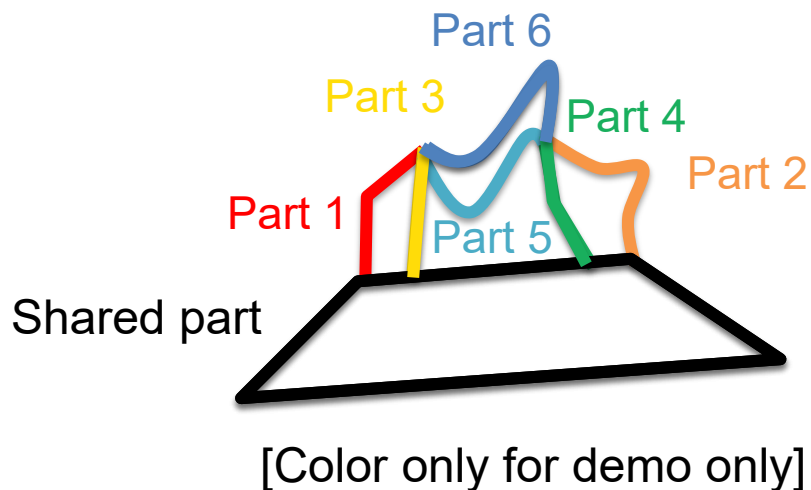
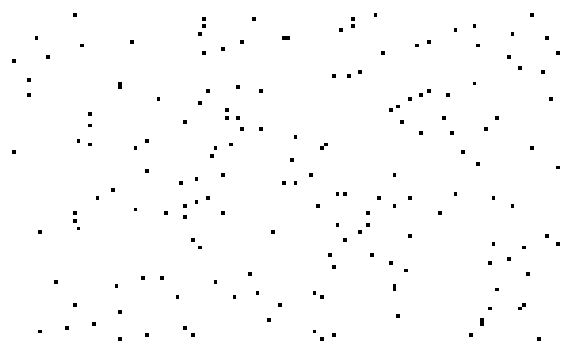


Image construction:



+Noise

Experiment training sets

Ownership matrix

Parts

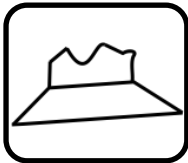


Img. 1	1	1	1	0	0	0
Img. 2	0	0	0	1	1	1

Img. 1



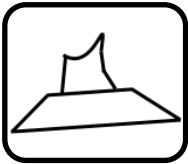
$f_1 =$



Img. 2

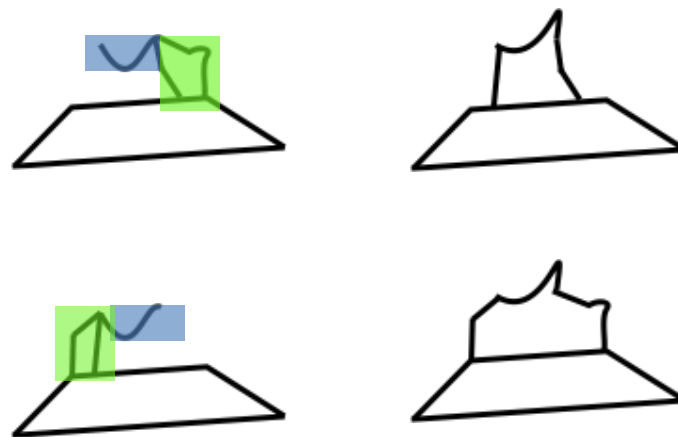


$f_2 =$



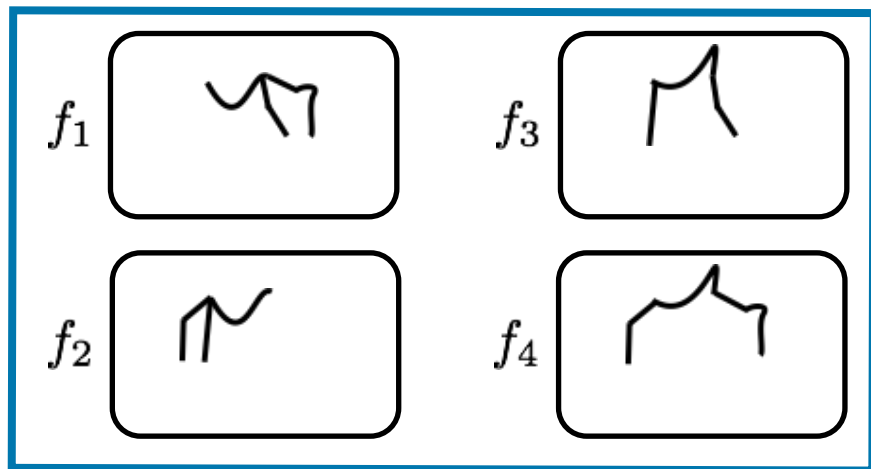
Experiment training sets

Images	Parts					
	✓	↗	↘	↖	↙	
	1	0	1	0	1	0
	1	0	0	1	0	1
	0	1	1	0	0	1
	0	1	0	1	1	0

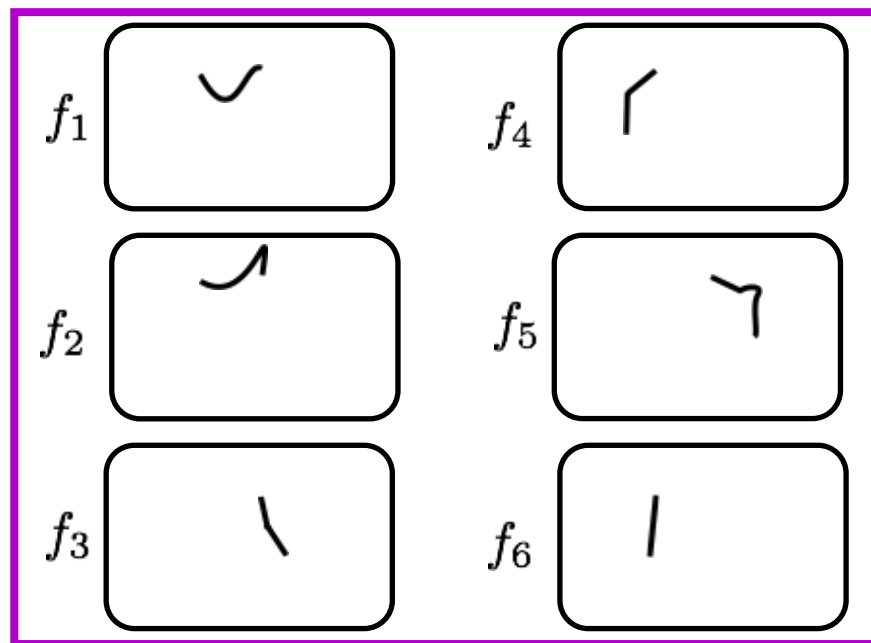


Experiment training sets

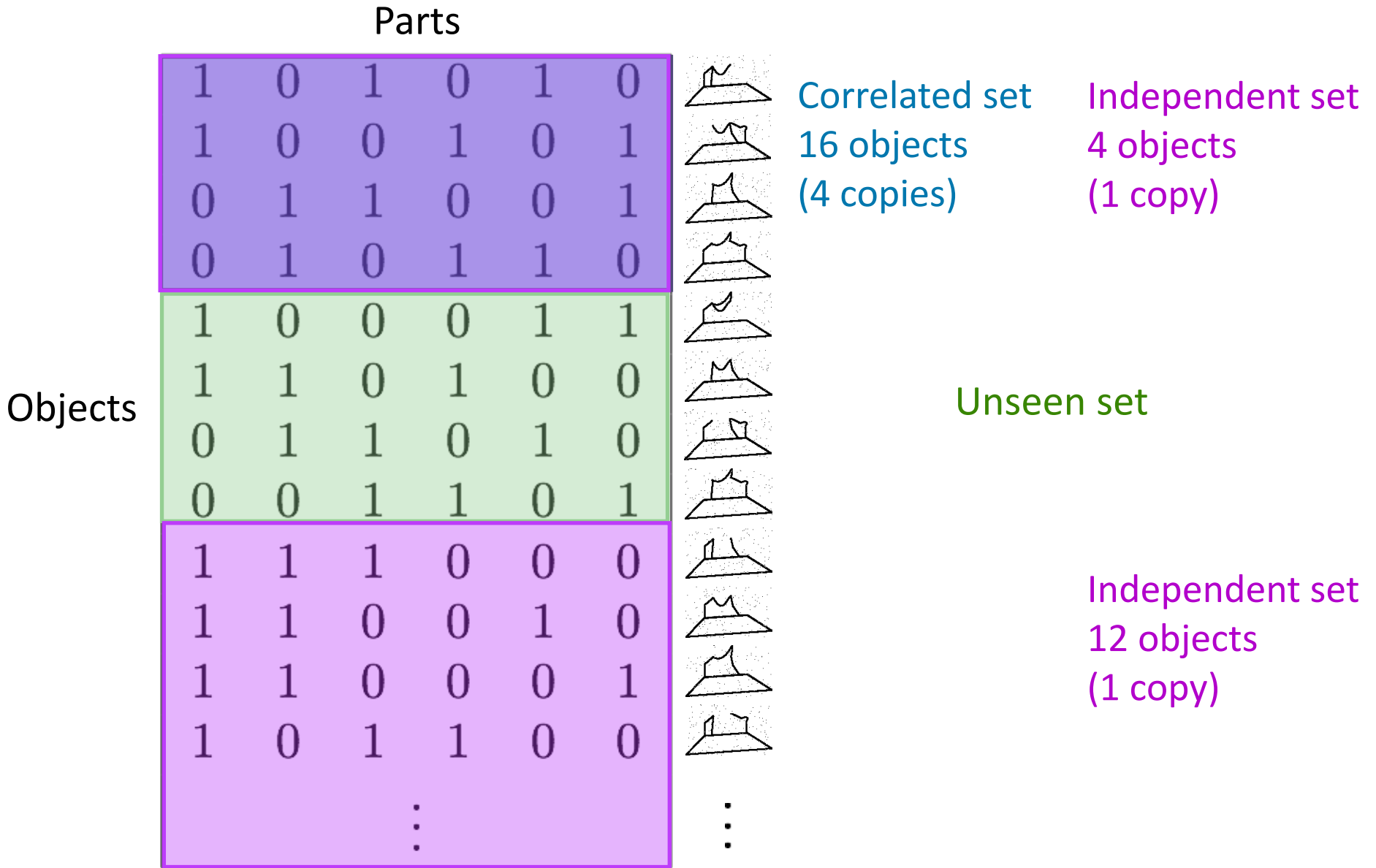
Correlated



Independent



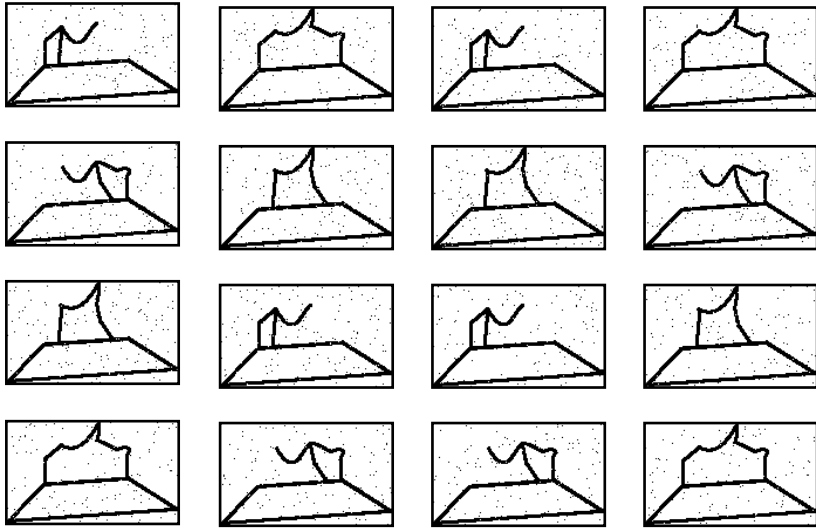
Experiment training sets



All 20 Possible Objects

Experiment procedure

1. Training phase: exploration



Two groups of participants:

Correlated

Independent

2. Testing phase: generalization



How likely do you think the rover is to see this image on another part of the cave wall?

Please circle the appropriate number:

0 1 2 3 4 5 6 7 8 9 10
not at all somewhat very
likely likely likely

Three types of images:

Seen

Unseen

Control (shuffled parts)

Computational framework

$$P\left(\text{Representations} \mid \text{Objects}\right) \propto P\left(\text{Objects} \mid \text{Representations}\right) P\left(\text{Representations}\right)$$

$P\left(\boxed{\parallel} = \text{Challenge: How to form a set over possible representations?}\right)$

$P\left(\left(\begin{array}{l} \boxed{\parallel} = f_1 \\ \boxed{\parallel} = f_2 \\ \boxed{\parallel} = f_3 \end{array}\right)\right)$ Use flexible hypothesis spaces from Bayesian nonparametrics.

Given a set of images, how should we represent it?

Encode images using two components:

Feature ownership matrix Z

		Features					
		✓	∟	∕	∖		⌢
Images	Img. 1	1	1	1	0	0	0
	Img. 2	0	0	0	1	1	1

Each element (image i , feature k) denotes whether image i has feature k .

Feature image matrix Y

		Pixels							
		1	...						D
Features	✓ 1	■	■	■	■	■	■	■	■
	∟	■	■	■	■	■	■	■	■
	∕	■	■	■	■	■	■	■	■
	∖	■	■	■	■	■	■	■	■

Encodes perceptual consequences of having each feature.

Computational framework

$$P(\mathbf{Z}, \mathbf{Y} | \mathbf{X}) \propto P(\mathbf{X} | \mathbf{Z}, \mathbf{Y}) \times P(\mathbf{Z}) \times P(\mathbf{Y})$$

Posterior
Representation
for the image set.

Likelihood recreates images
despite noise.

$P(\text{image pixel})$ grows with the
number of features that have
the pixel.

Images composed via OR
(Pearl, 2000; Wood, et al., 2006).

Feature ownership prior
encodes expectations
about feature ownership.

*Biased towards fewer
features.*

Use the Indian buffet
process, a Bayesian
nonparametric process
(Griffiths & Ghahramani, 2005).

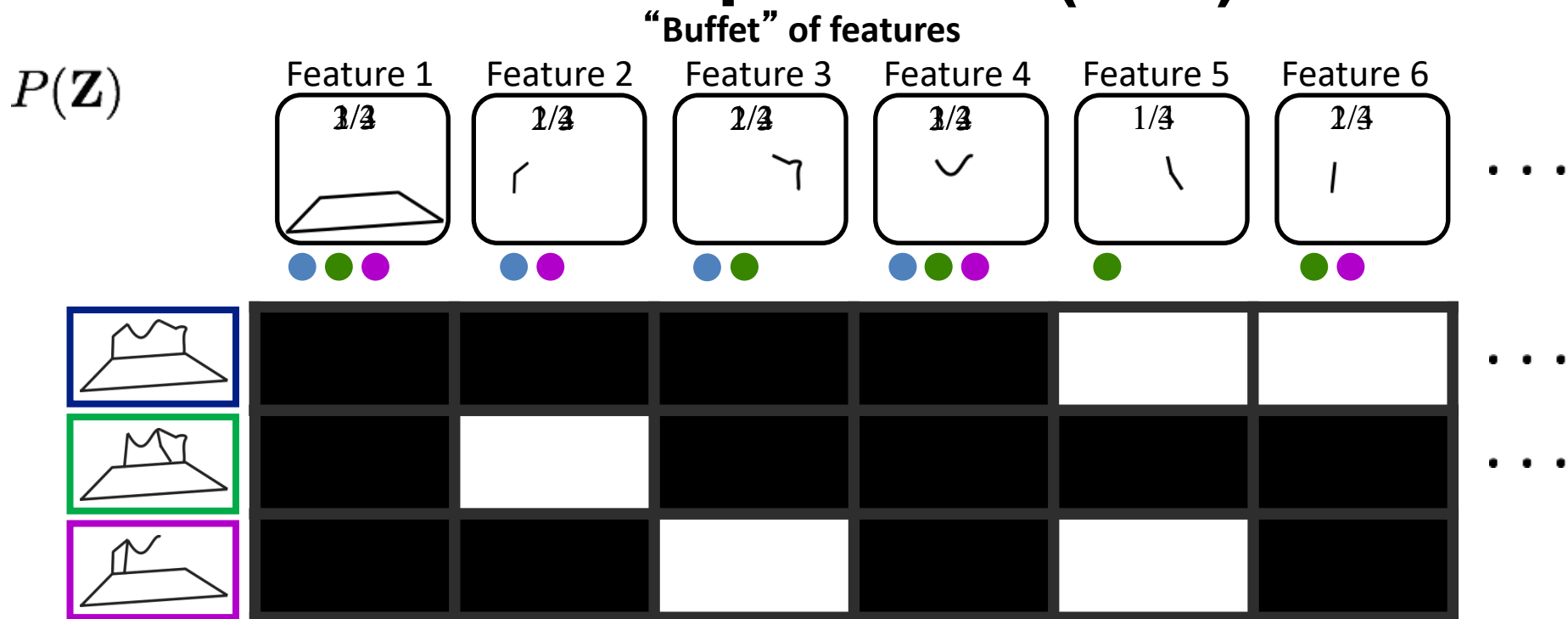
Feature image prior
encodes expectations
about feature images.

Pixel independence

Or use an Ising model
(Geman & Geman, 1984).

**Use standard machine learning techniques to infer the best
representation for an image set, once probabilities are specified.**

The Indian buffet process (IBP)



N customers (objects) take dishes (features) from an Indian buffet....

customer i takes “Old” features

$$m_k/i$$

“New” features

$$\text{Poisson}(\alpha/i)$$

Customer 1 (takes 4 dishes):

N/A

Samples 4

Customer 2 (takes 5 dishes):

1,3,4

Samples 2

Customer 3 (takes 4 dishes):

1,2,4,6

Samples 0

old feature term

$$P(\mathbf{Z}) \propto \left(\prod_{k=1}^{K_+} \frac{(N - m_k)!(m_k - 1)!}{N!} \right) \exp \{ -\alpha H_N \} \alpha^{K_+}$$

new feature term

Griffiths & Ghahramani (2005)

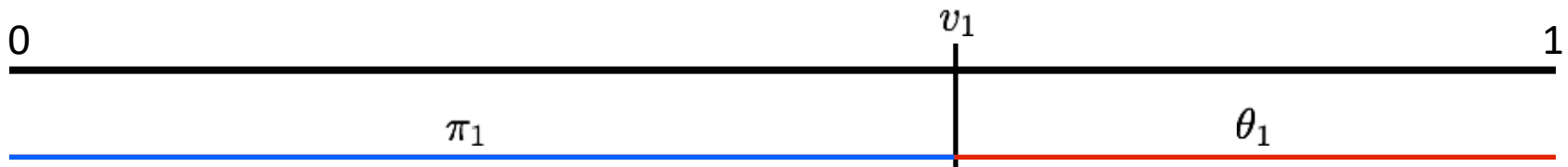
Stick-breaking for the CRP and IBP

A process that generates the CRP and IBP, but does not integrate out the latent weights:

$$v_k \stackrel{iid}{\sim} \text{Beta}(\alpha, 1)$$

$$\text{CRP} \quad \theta_k = (1 - v_k) \prod_{l=1}^{k-1} v_l$$

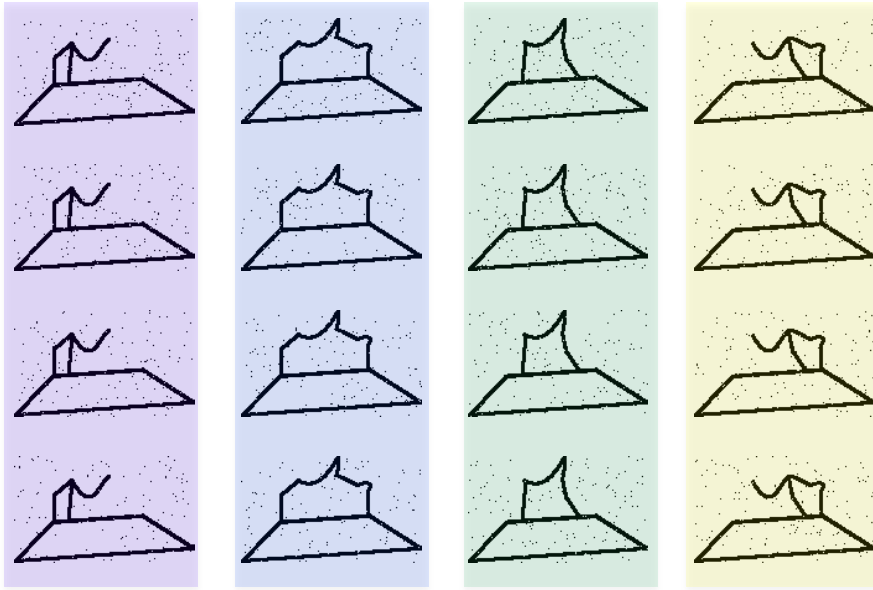
$$\text{IBP} \quad \pi_k = v_k \prod_{l=1}^k v_l$$



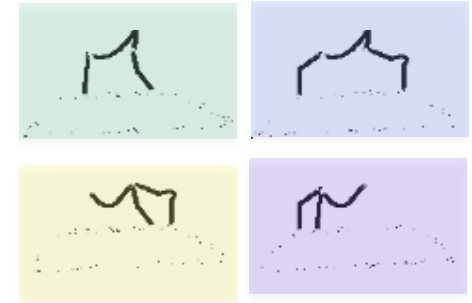
Although $\vec{\pi}$ is strictly decreasing, $\vec{\theta}$ is not (in the example, $\theta_3 > \theta_2$)!

Distribution on $\vec{\theta}$ is also called the Griffiths-Engel-McCloskey (GEM) distribution.

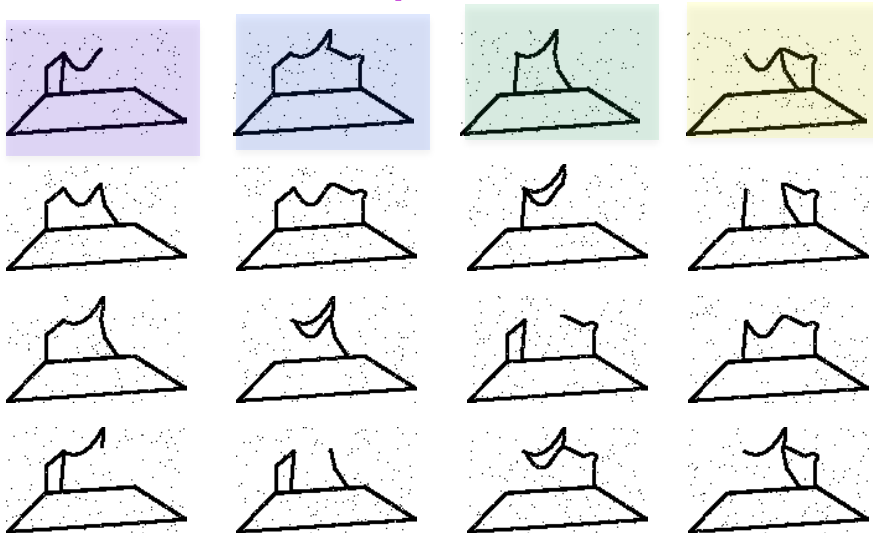
Correlated



IBP Inferred Features

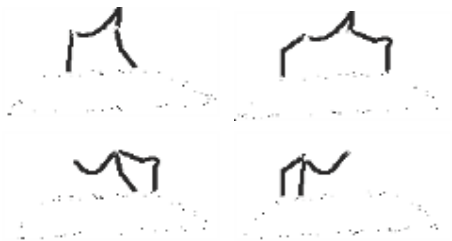


Independent

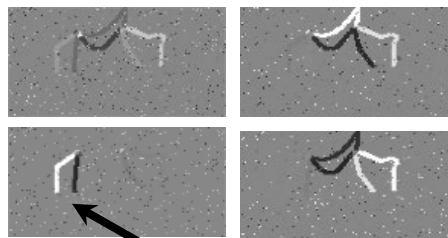


Correlated

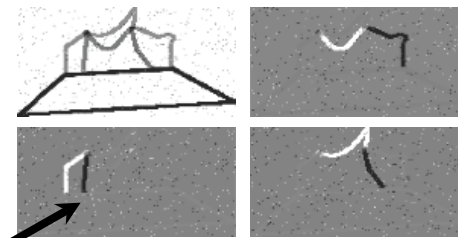
IBP Features



PCA Features

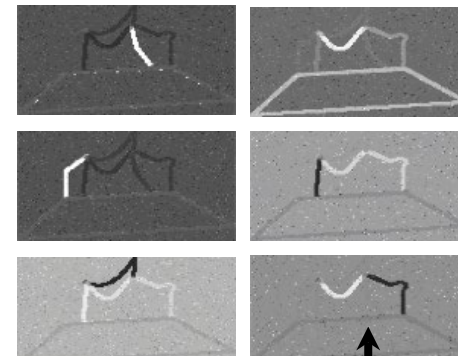
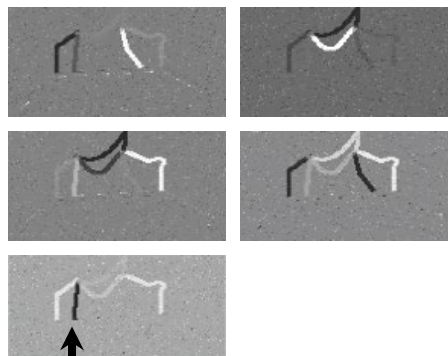
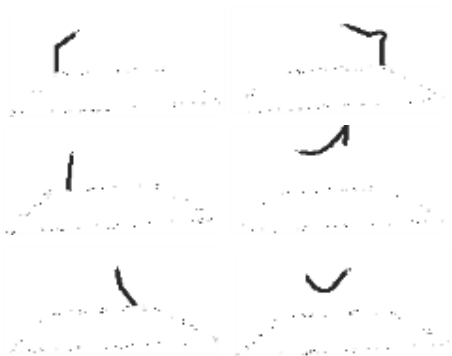


ICA Features



Parts 1 & 2 are negatively correlated.

Independent



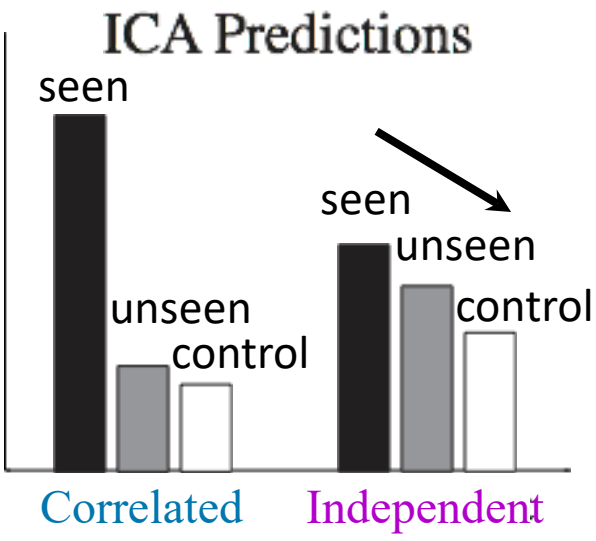
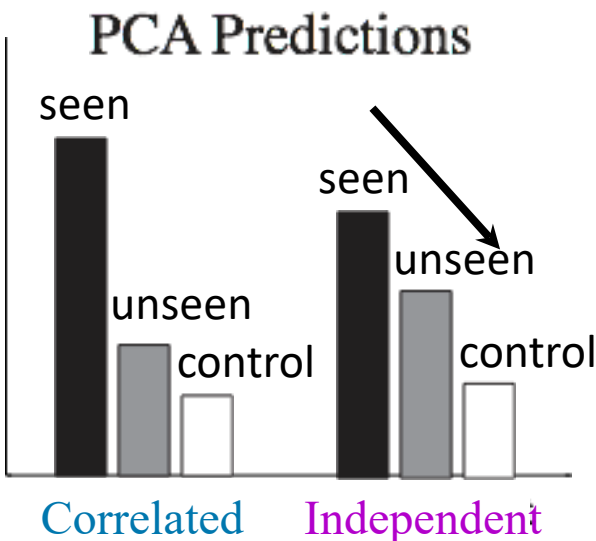
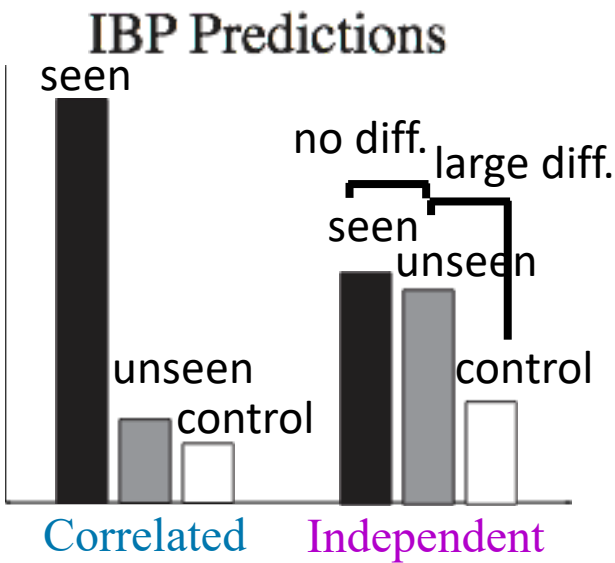
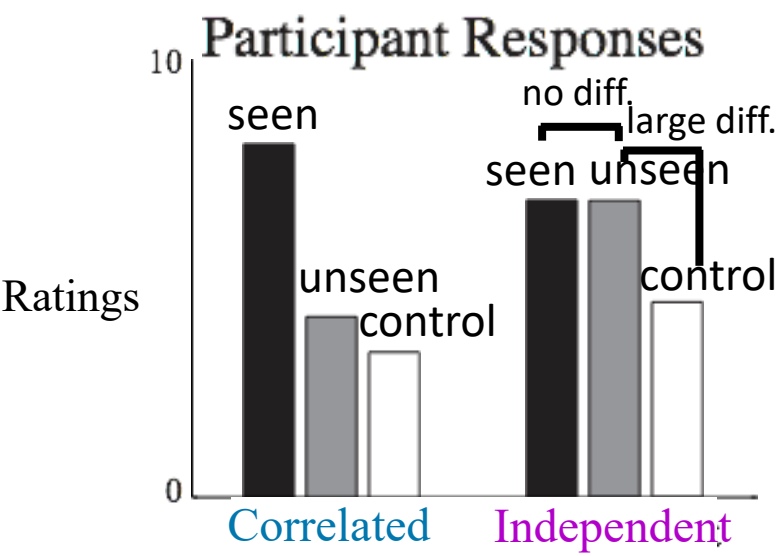
Parts 1 & 2 negatively are correlated.

Parts 4 & 5 negatively are correlated.

What do they predict?

Results

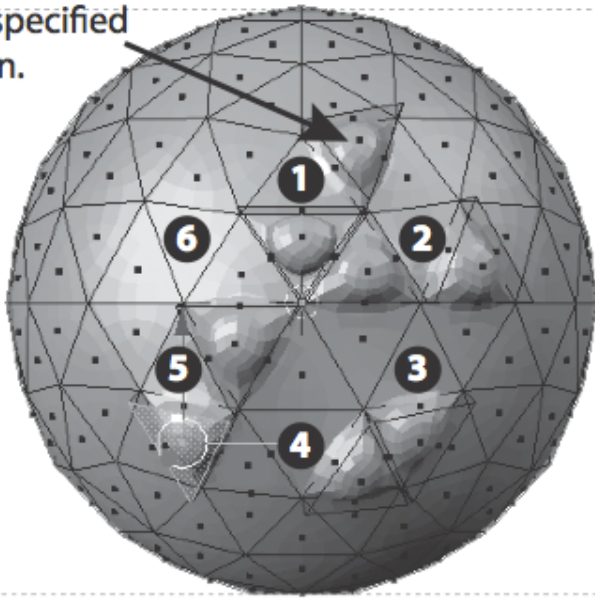
All models only have one parameter fit to participant results.
IBP also outperforms exemplar and prototype models.



Domain generality?

3-D Rendered Objects

A part is a "bump"
at the specified
location.



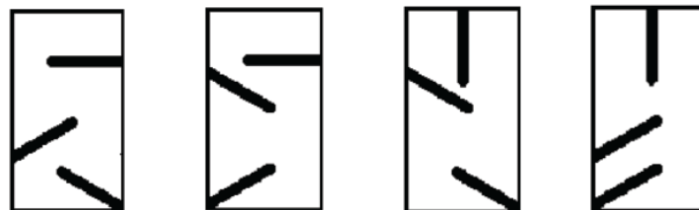
Each object has the bumps shown above and
3 bumps at 1 of the 6 locations.

Concepts

- (1) Lays eggs.
- (2) Moves fast.
- (3) Has small claws.
- (4) Spends most time on land.
- (5) Has scaly skin.
- (6) Is a small herbivore.

Capturing prior expectations: contiguity

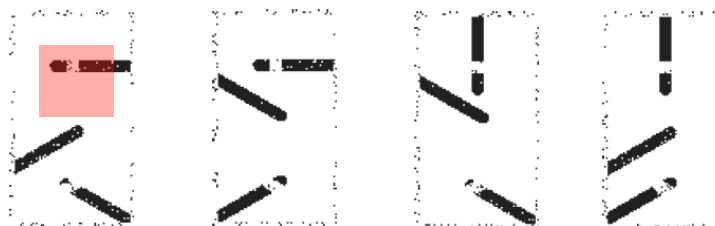
Visual search for objects with correlated parts
(Shiffrin & Lightfoot, 1997).



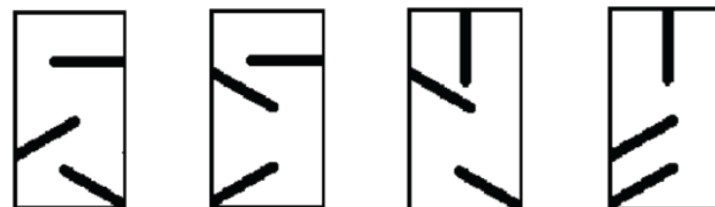
Features inferred without proximity constraint.

$$P(\mathbf{Y}) = \prod_{k,d} \phi^{y_{kd}} (1 - \phi)^{1-y_{kd}}$$

coin flip bias



Features inferred with proximity constraint.

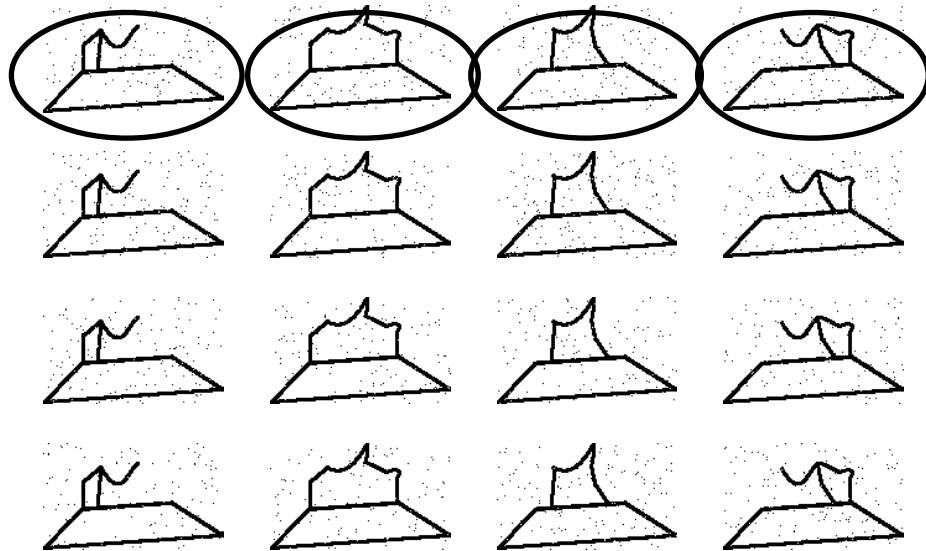


$$P(\mathbf{Y}) = \left(\prod_{k,d} \phi^{y_{kd}} (1 - \phi)^{1-y_{kd}} \right) \left(\prod_{\{d_1, d_2\} \in C} \boxed{\rho^{d_1 d_2} (1 - \rho)^{1-d_1 d_2}} \right)$$

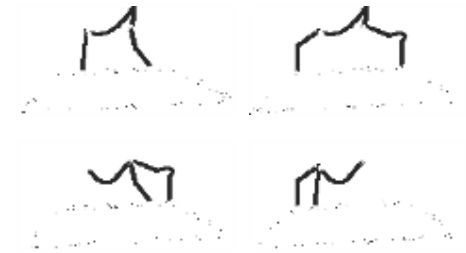
adj. pixels share value

Austerweil & Griffiths (2011)

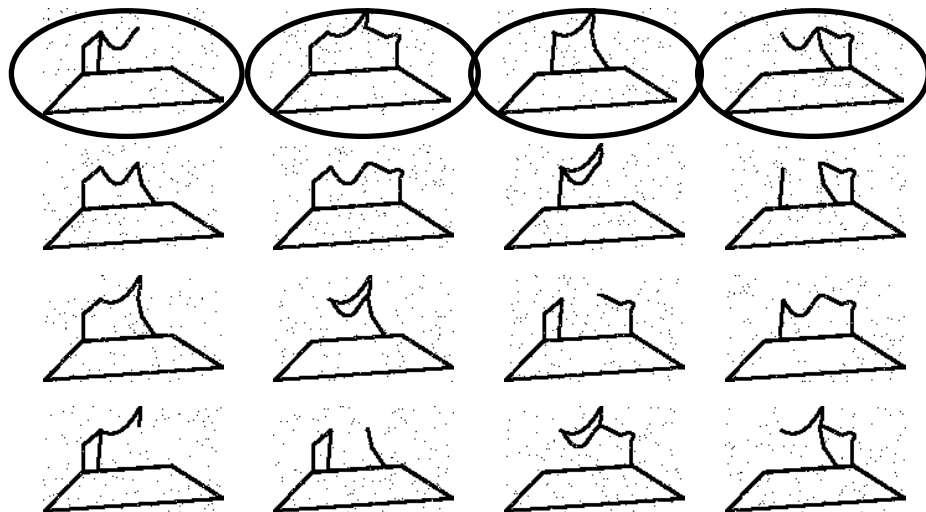
Correlated



IBP Inferred Features



Independent

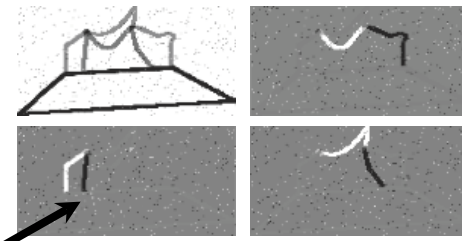
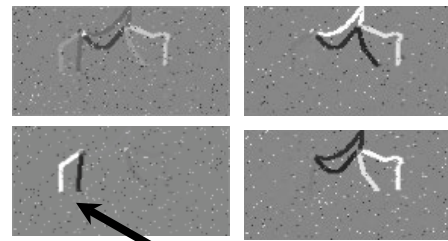
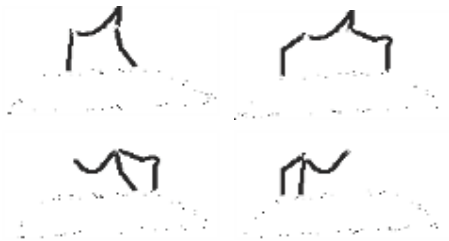


IBP Features

PCA Features

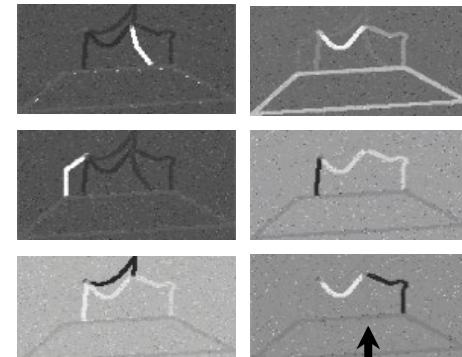
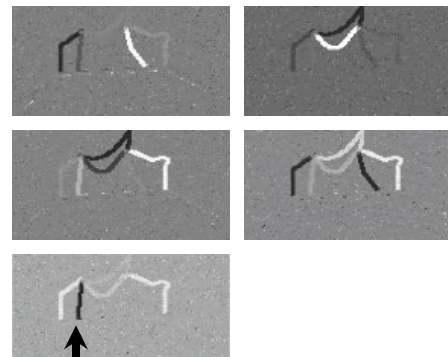
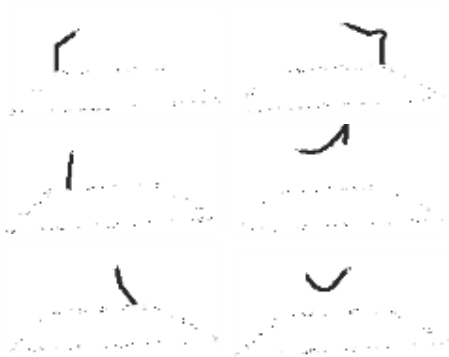
ICA Features

Correlated



Parts 1 & 2 are negatively correlated.

Independent



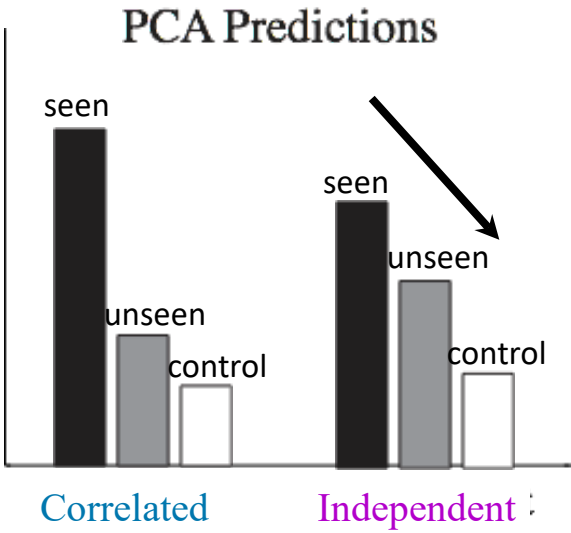
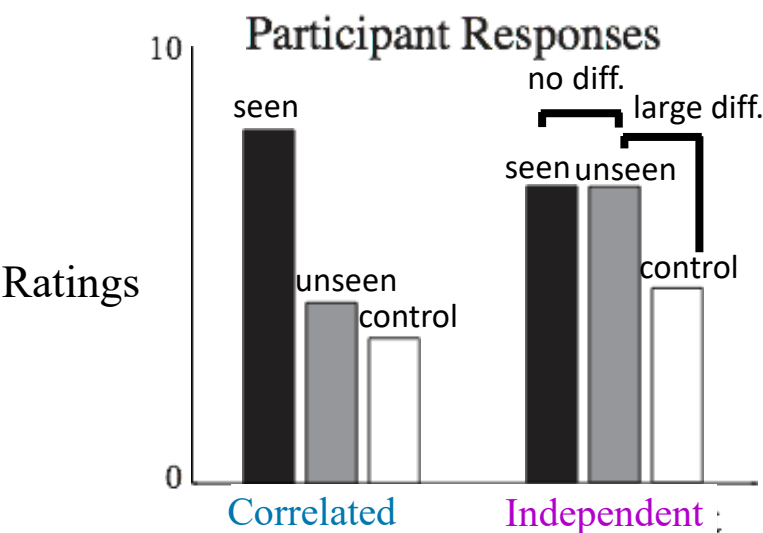
Parts 1 & 2 negatively are correlated.

Parts 4 & 5 negatively are correlated.

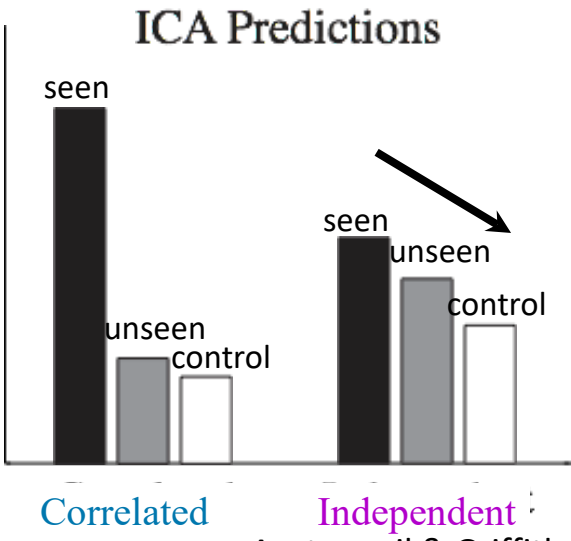
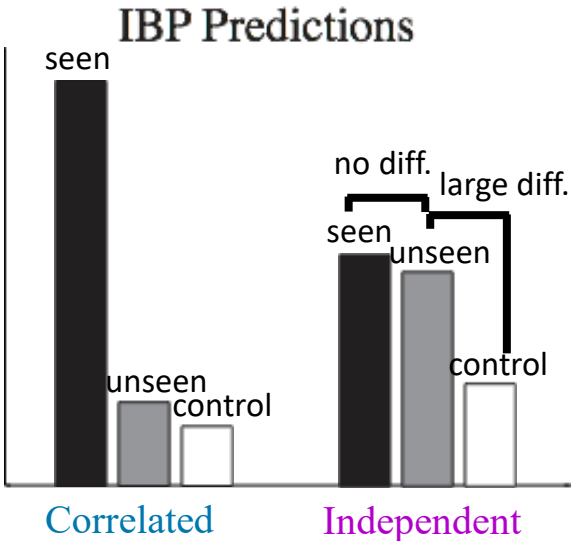
What do they predict?

Results

All models only have one parameter fit to participant results.



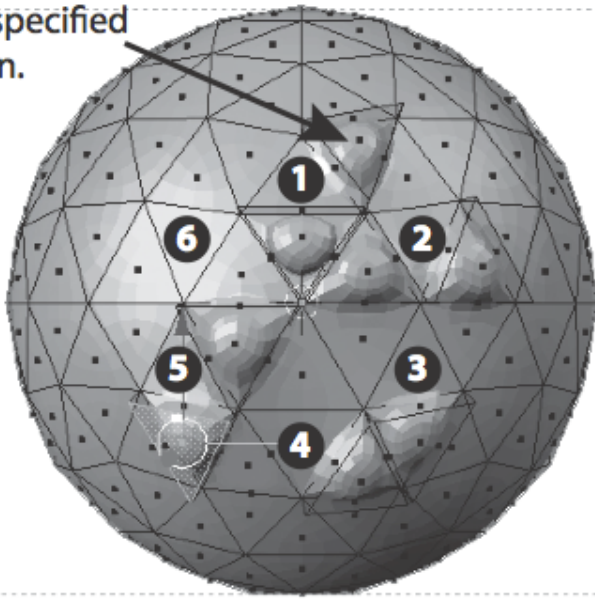
Given raw pixel data of images, the IBP infers the same features as people do, but PCA and ICA do not.



Domain generality?

3-D Rendered Objects

A part is a "bump"
at the specified
location.



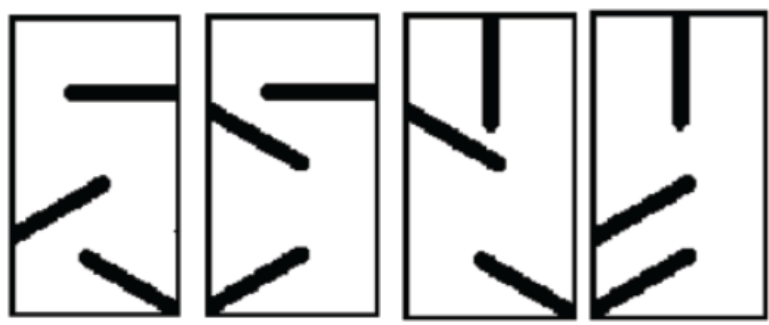
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3 bumps at 1 of the 6 locations.

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Visual search for objects with correlated parts (Shiffrin & Lightfoot, 1997).



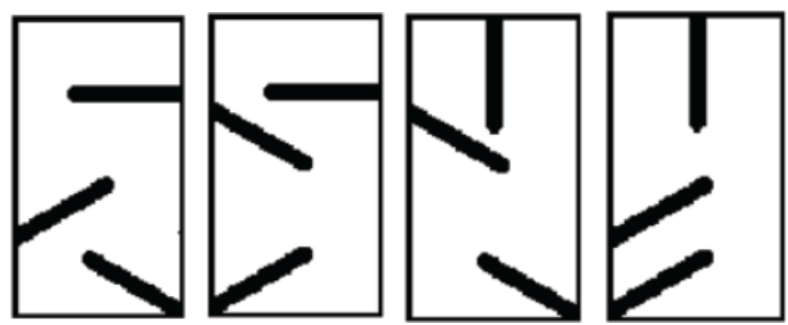
Features inferred without proximity constraint.

$$P(\mathbf{Y}) = \prod_{k,d} \boxed{\phi^{y_{kd}} (1 - \phi)^{1-y_{kd}}}$$

coin flip bias



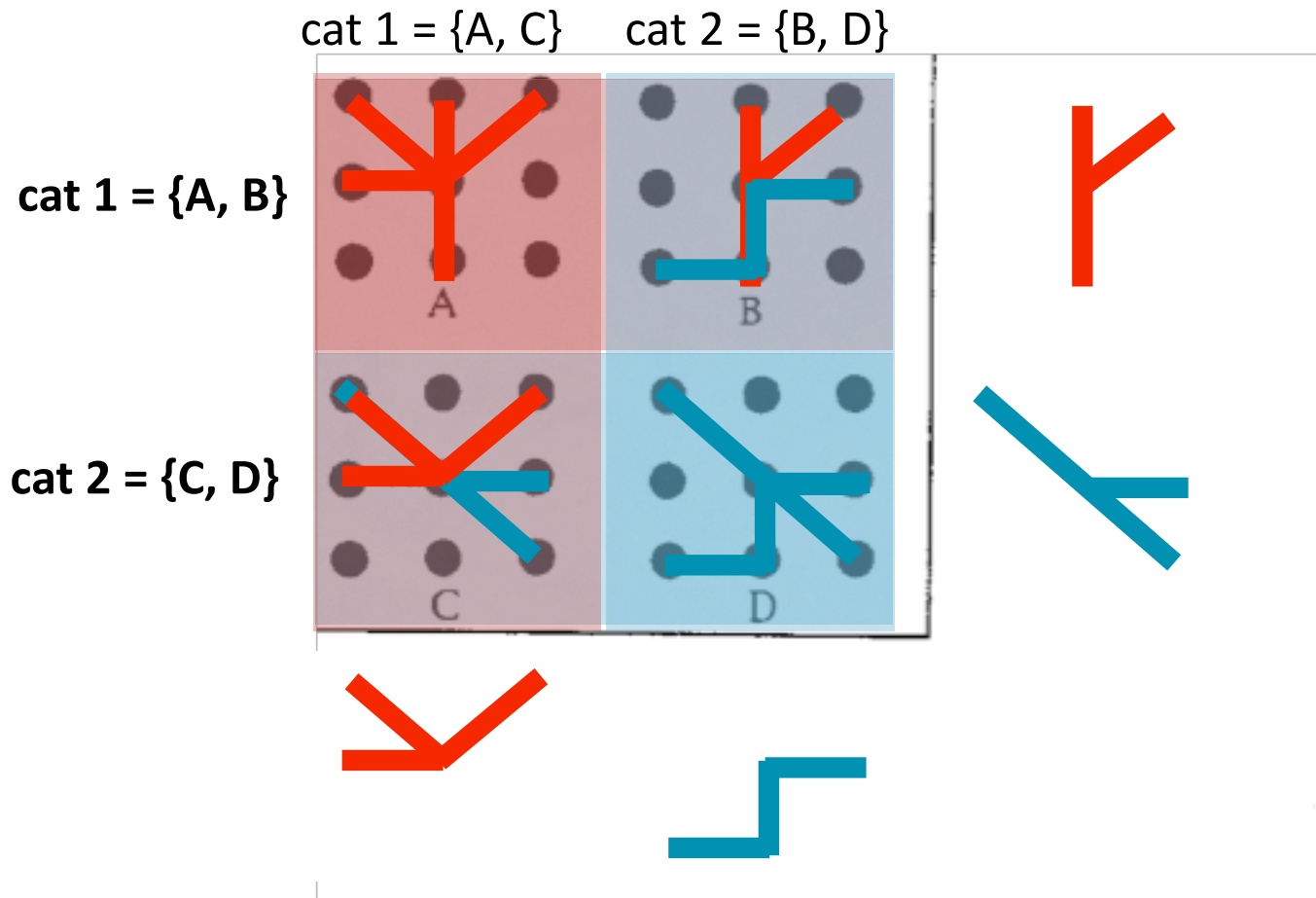
Features inferred with proximity constraint.



$$P(\mathbf{Y}) = \left(\prod_{k,d} \phi^{y_{kd}} (1 - \phi)^{1-y_{kd}} \right) \left(\prod_{\{d_1,d_2\} \in C} \boxed{\rho^{d_1 d_2} (1 - \rho)^{1-d_1 d_2}} \right)$$

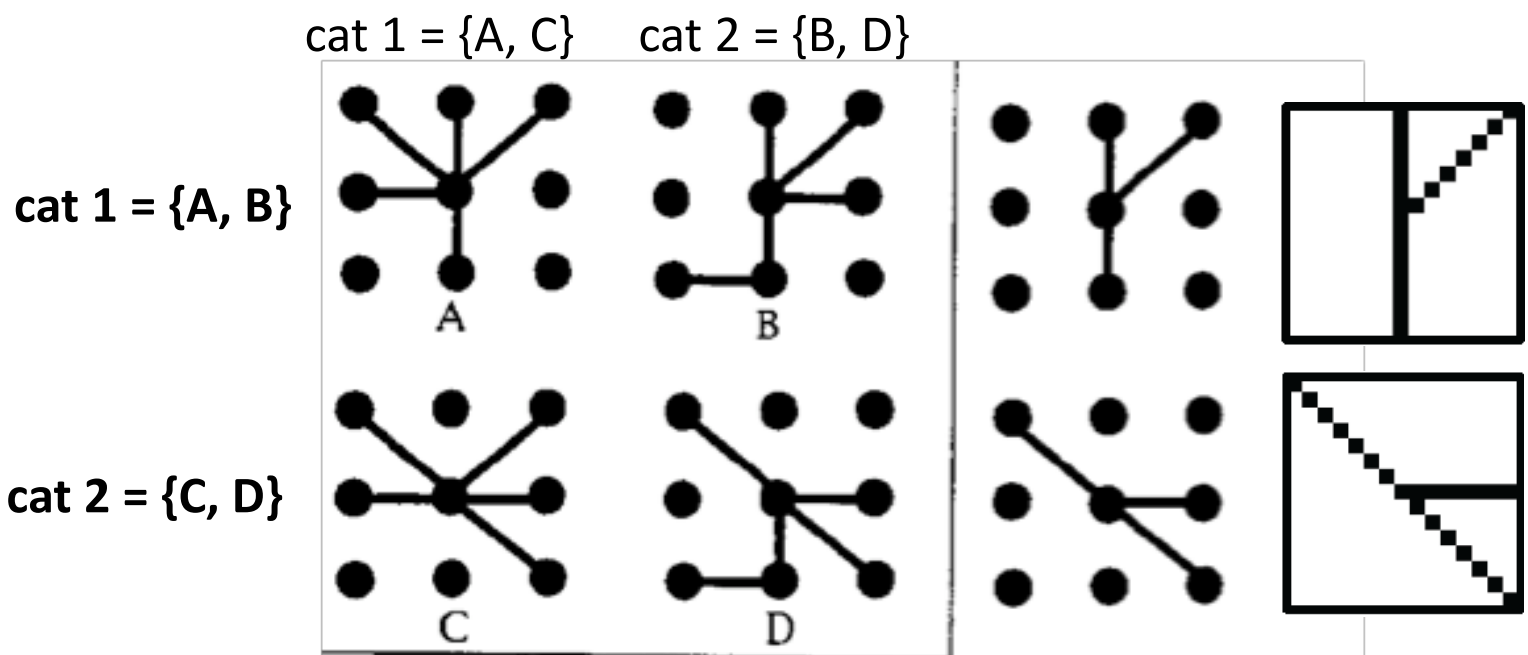
adj. pixels share value Austerweil & Griffiths (2009; 2011)

Category diagnosticity



Task: Object categorization with feedback
Result: Participants learn features appropriate to their training condition.

Category diagnosticity



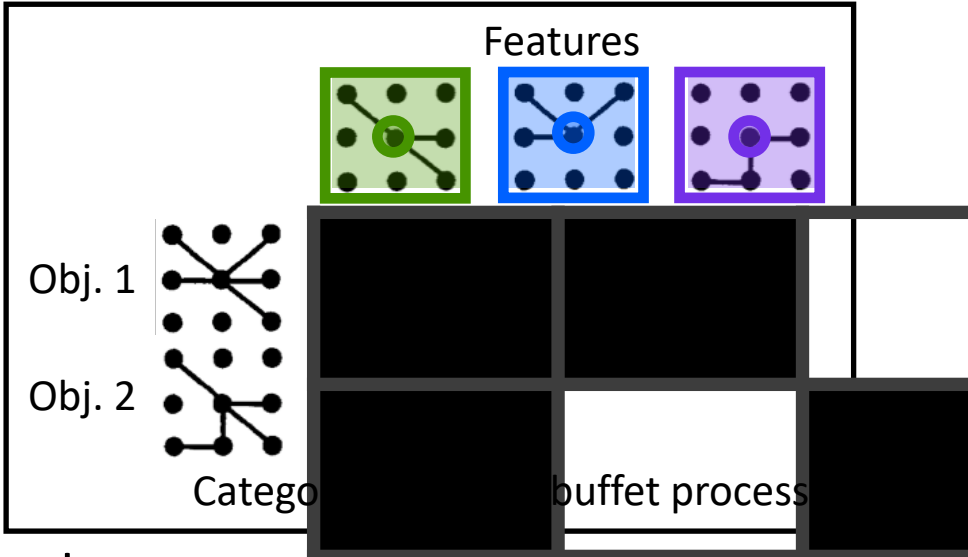
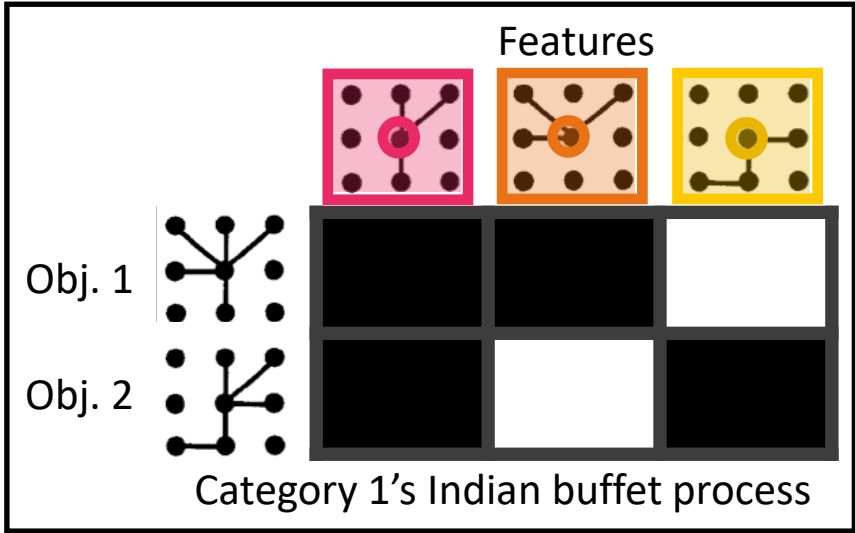
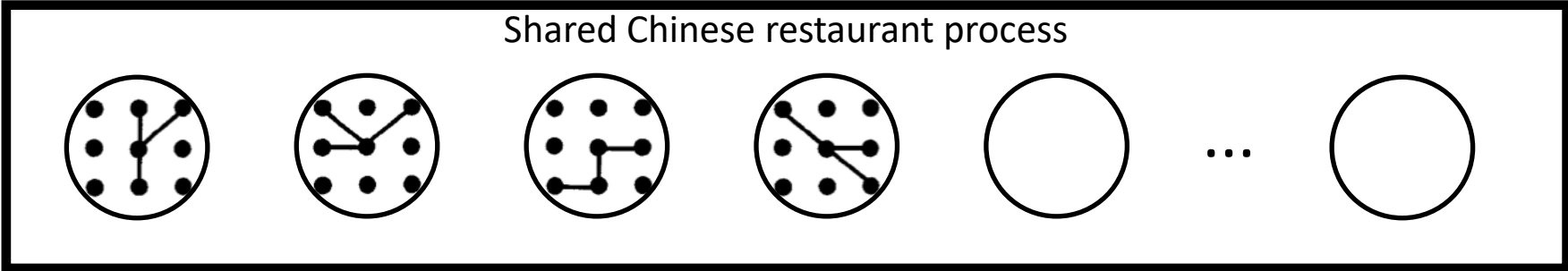
Include category label as part of an object's "sensory inputs."

... but, only representing an object with its diagnostic feature is not enough!

How else can categorization influence the features inferred by a model?

The Indian buffet franchise

(Austerweil & Griffiths, in rev.)

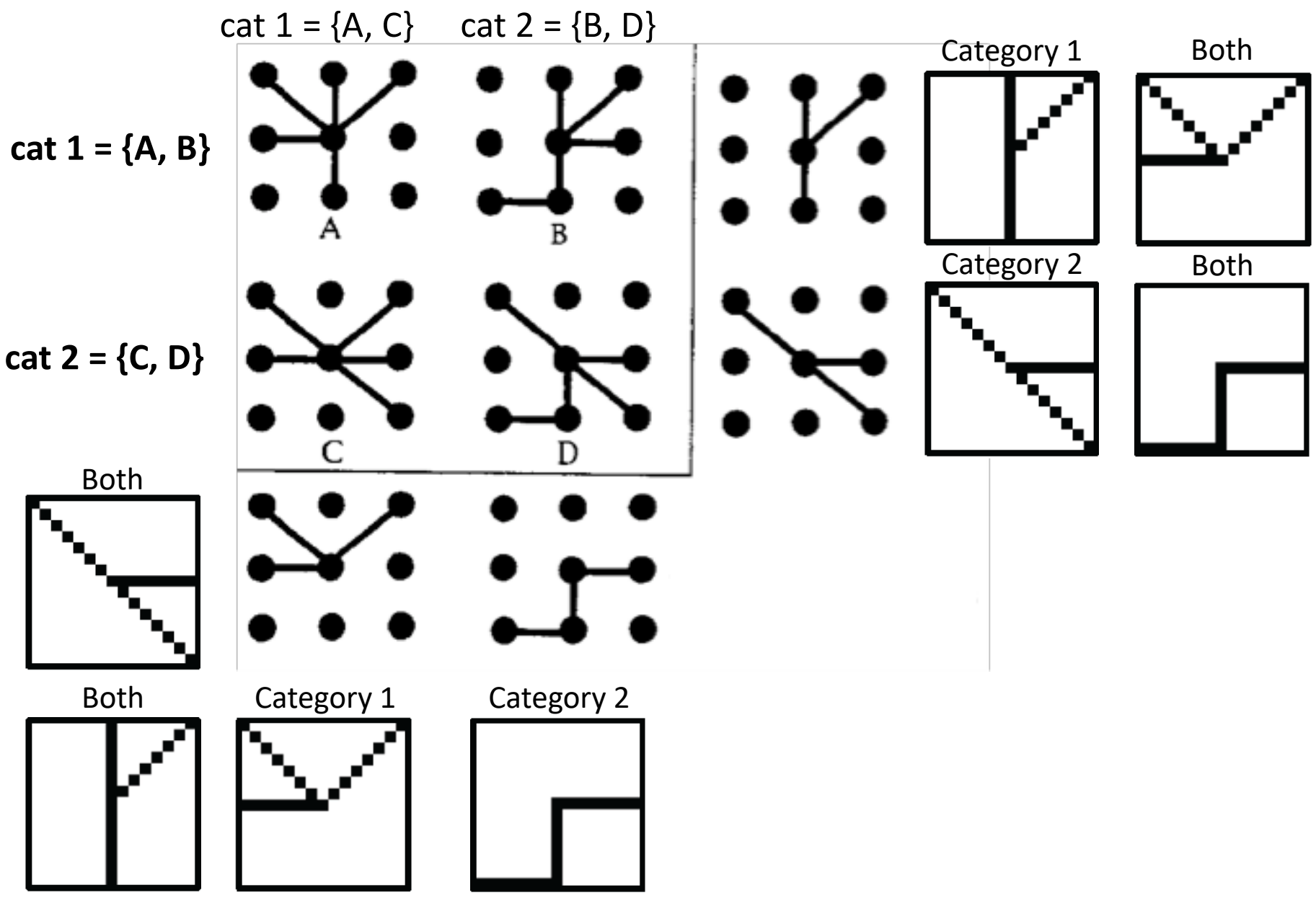


... But should feature images be shared across categories? If so, how?

Assume that the feature images come from a common source:
a Chinese restaurant process (Aldous, 1983).

This method for sharing information across categories is used to model human learning of hierarchical categories (Griffiths et al., 2011).

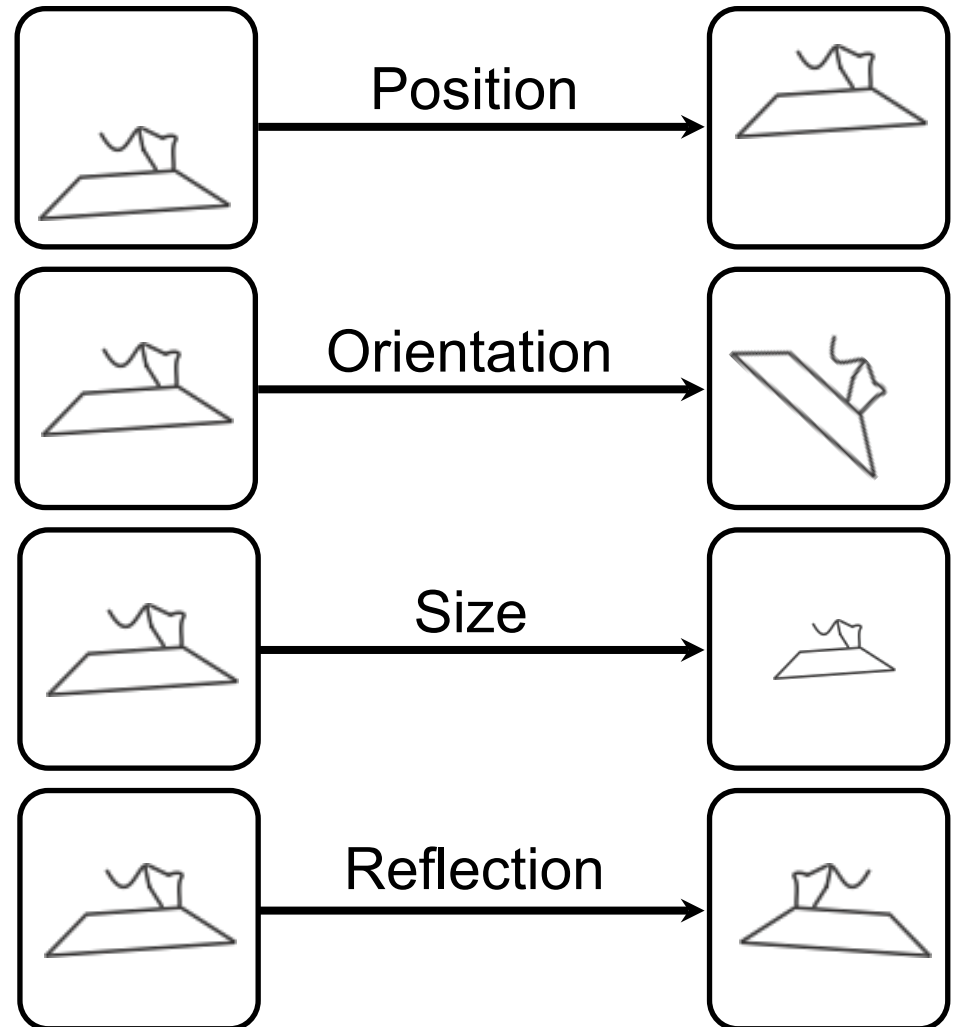
Category diagnosticity



Learning transform invariant features

Previous work focused on inferring features that occur the same way across presentations.

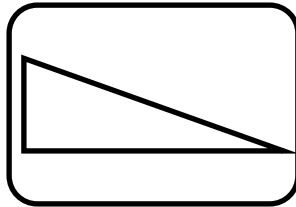
However, features can be *transformed* across presentations (Palmer, 1983).



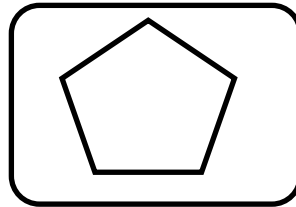
The transformed Indian buffet process

“Buffet” of features

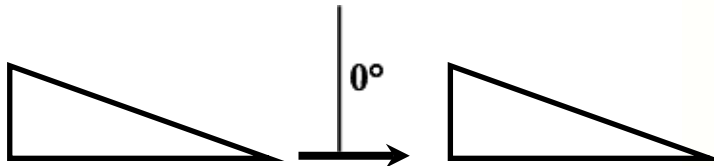
Feature 1



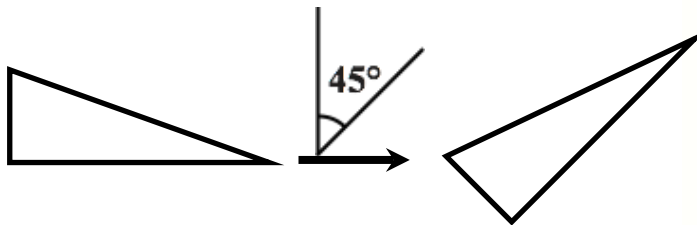
Feature 2



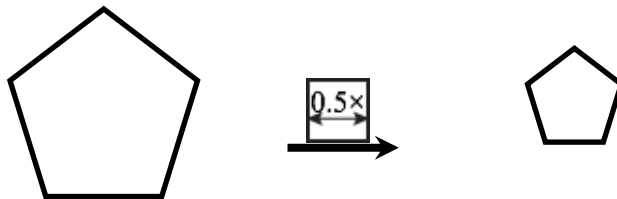
• $\mathbf{x}_1$:



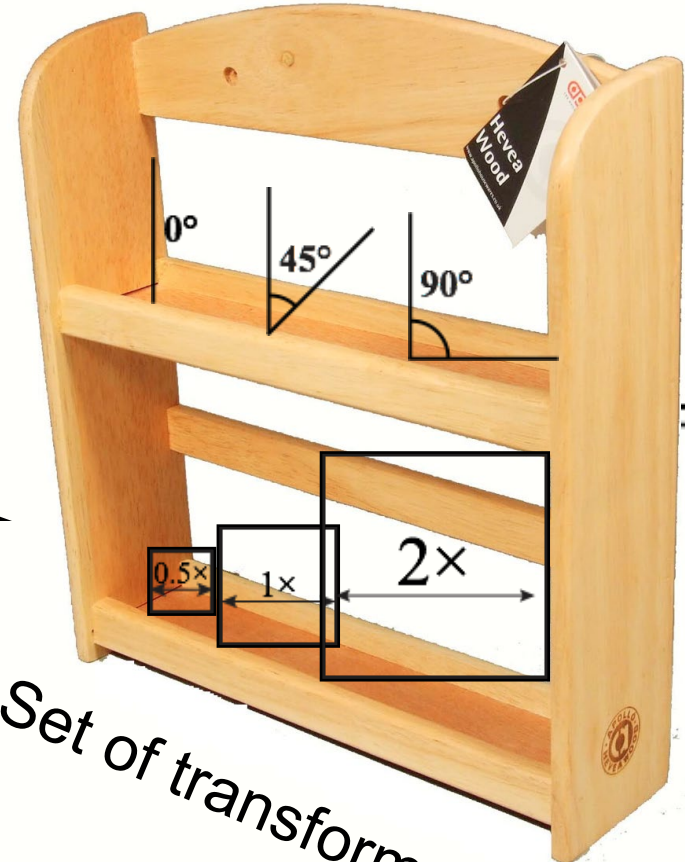
• $\mathbf{x}_2$:



• $\mathbf{x}_3$:



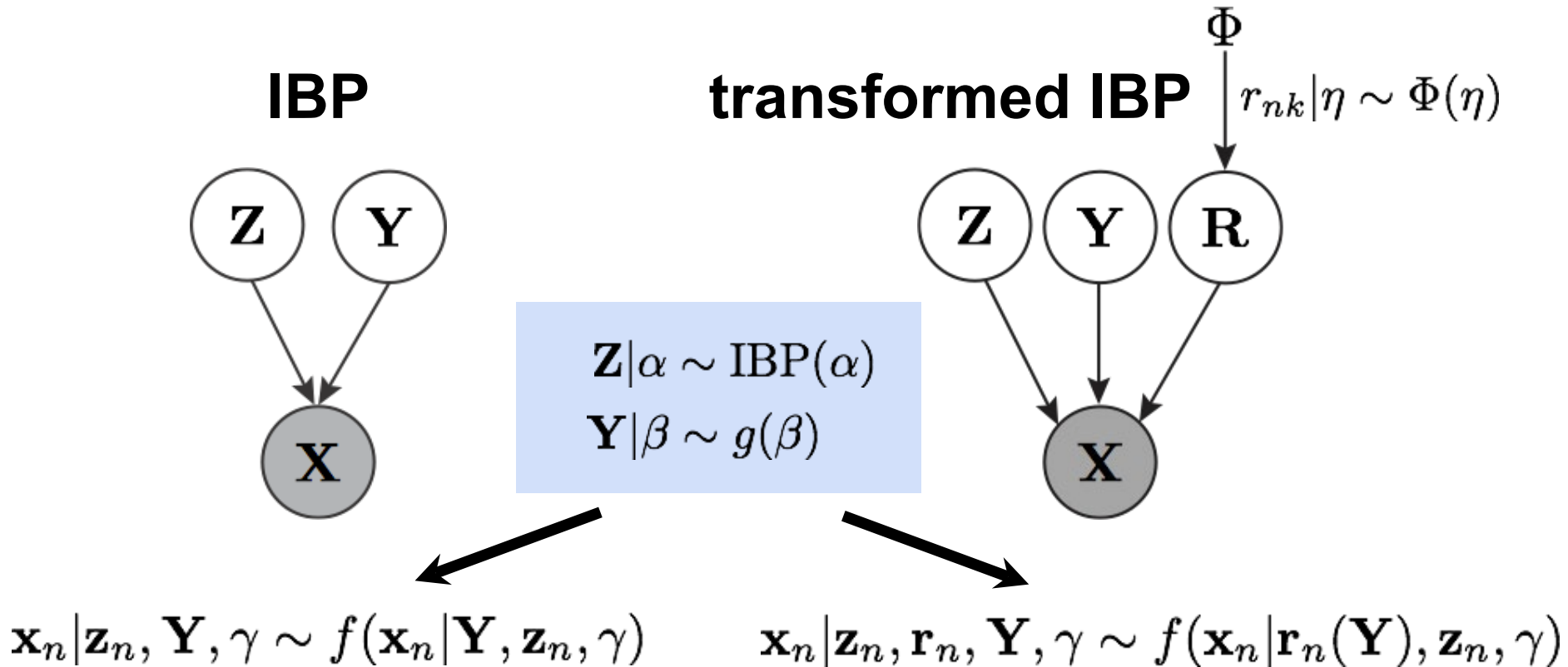
Spice rack



$= \Phi$

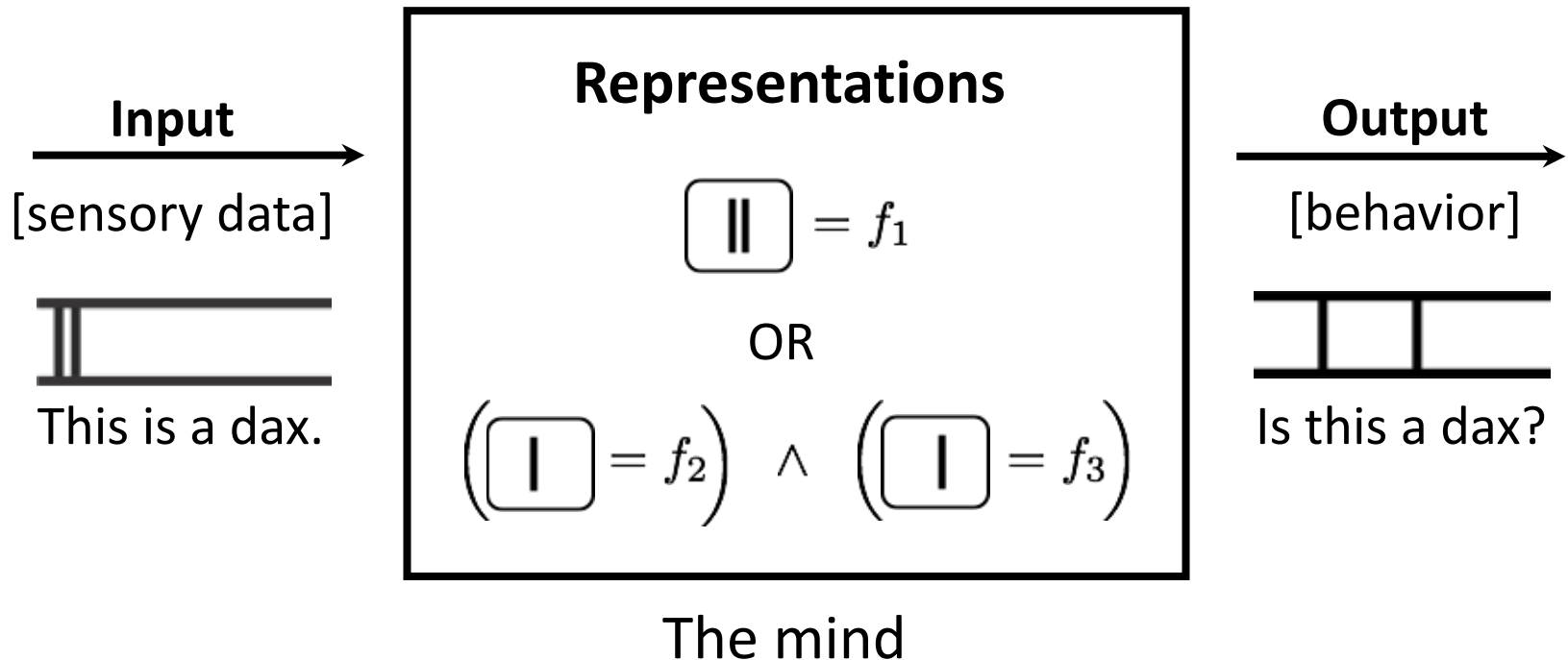
Set of transformations

The transformed Indian buffet process



Learning transformation-invariant features

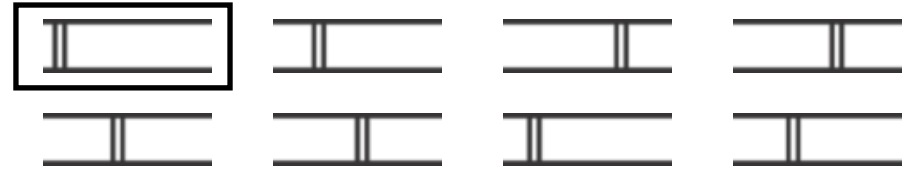
Features occur differently across presentations.
Are parts separate features or the same feature with different transformations?



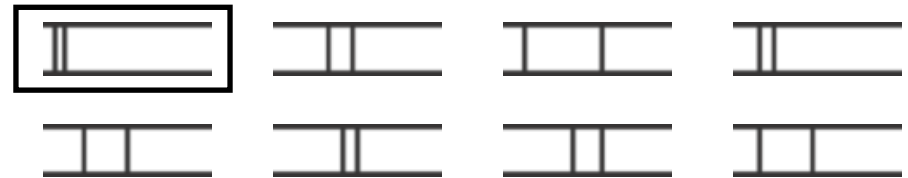
Experiment Stimuli

Two object sets

Unitized: vertical bars translated together



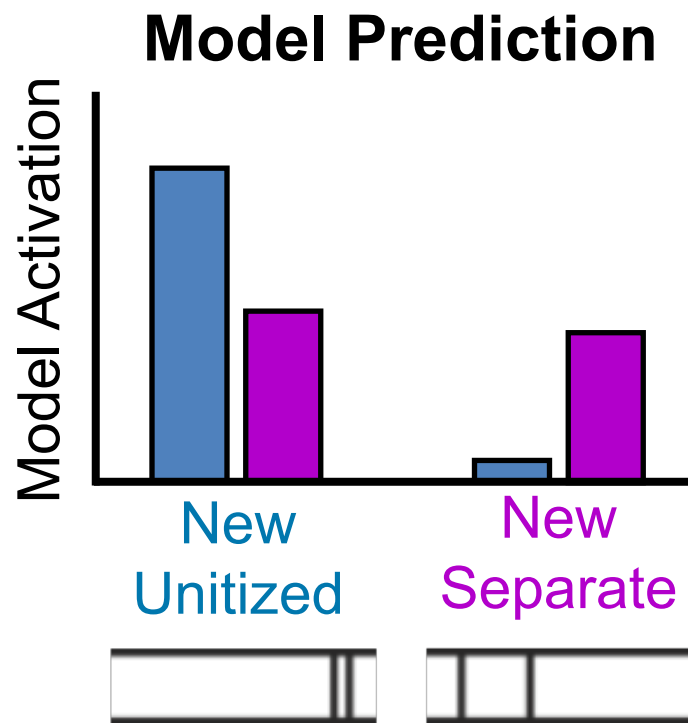
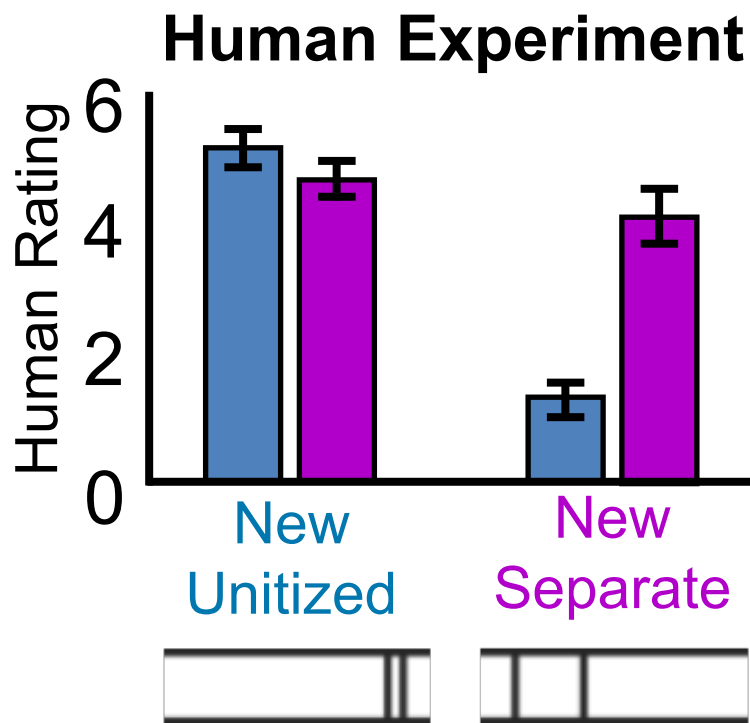
Separate: vertical bars translated independently



Hypothesis

People infer the representation that encodes the observed objects with the fewest features.

Learning transformation-invariant features



Unitized



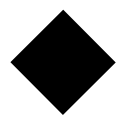
Separate



Learning feature transformation structure

Should all transformations be included? (More invariance the better?)

5 5 d 5

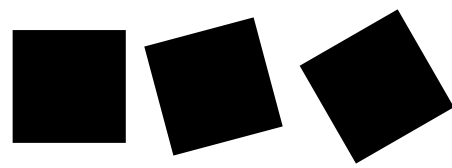
  Square or diamond?
(Mach, 1914)

Hypothesis: people infer which transformations a feature undergoes.

Training Set

Test Set

Rotation

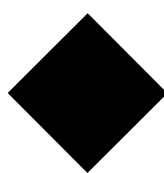


or

Size



New Rotation



New Size



Learning feature transformation structure

Training Set

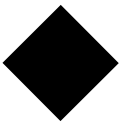
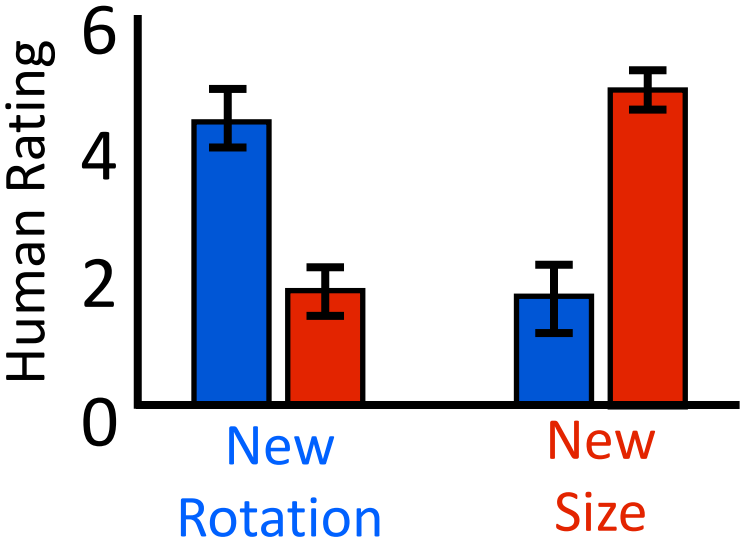
Rotation



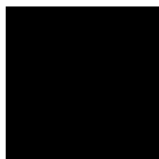
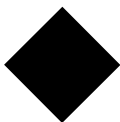
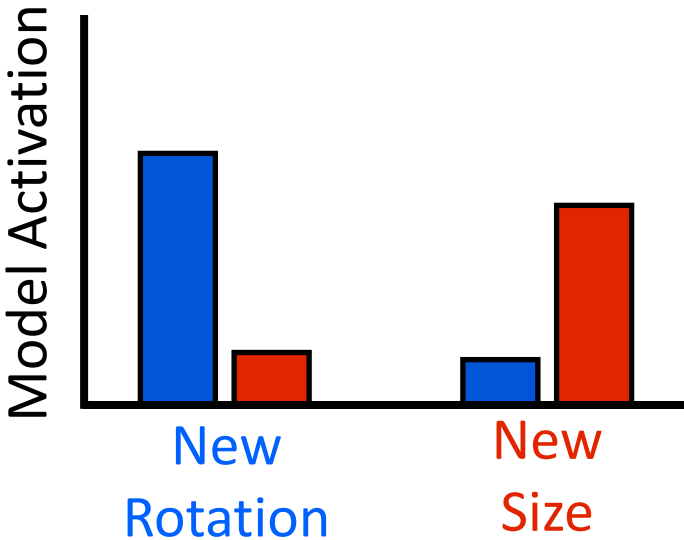
Size



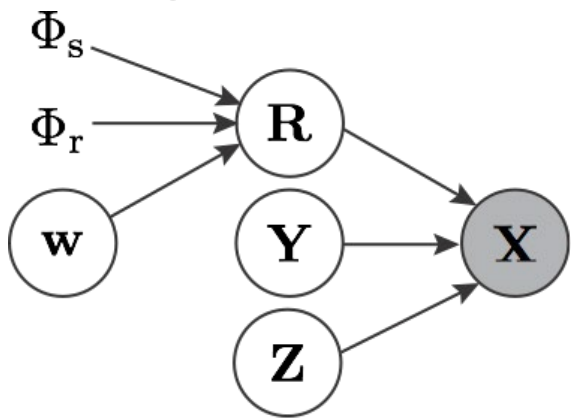
Human Experiment



Model Prediction



Learning feature transformation structure



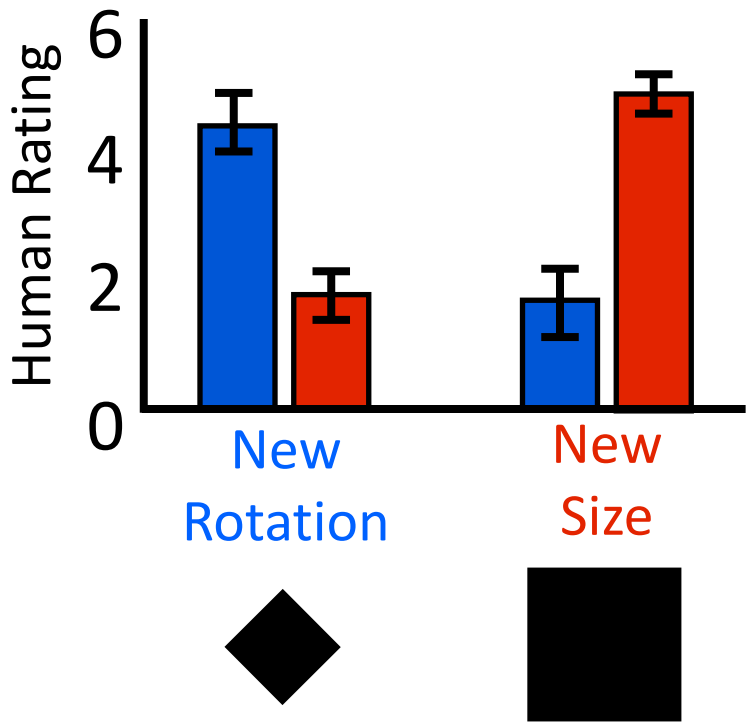
$$\mathbf{w} = (w_k)_{k=1}^{\infty}, w_k \in \{0, 1\}$$

Indicates type of transformation for feature k .

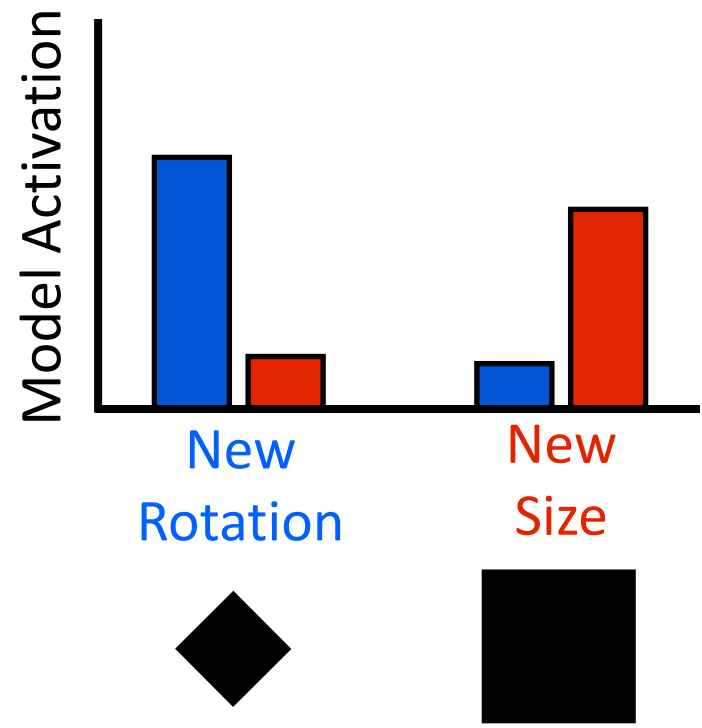
Φ_r is a prior on rotation transformations.

Φ_s is a prior on scaling transformations.

Human Experiment



Model Prediction



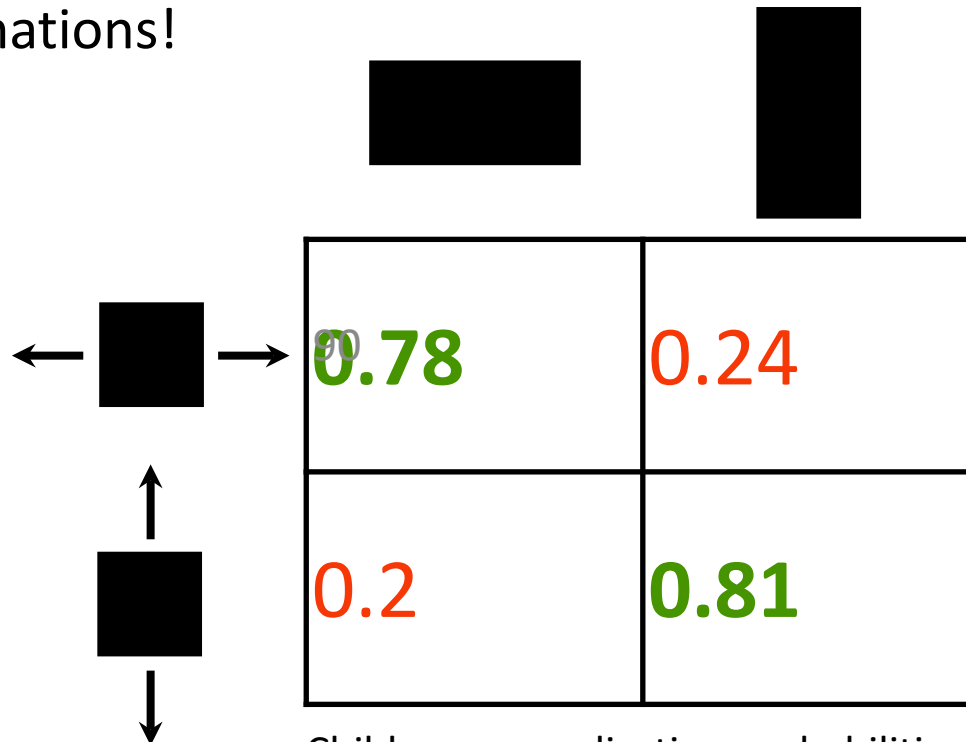
Learning feature transformation structure

Sometimes information is shared between the types of transformations.

Smith (2005) taught children the name of a new object, which either moved *horizontally* or *vertically*.

Children generalized to novel objects that were scaled in the same direction that the original object was moved.

Information about translation transformations was shared with scaling transformations!



Children generalization probabilities Austerweil & Griffiths (in revision)

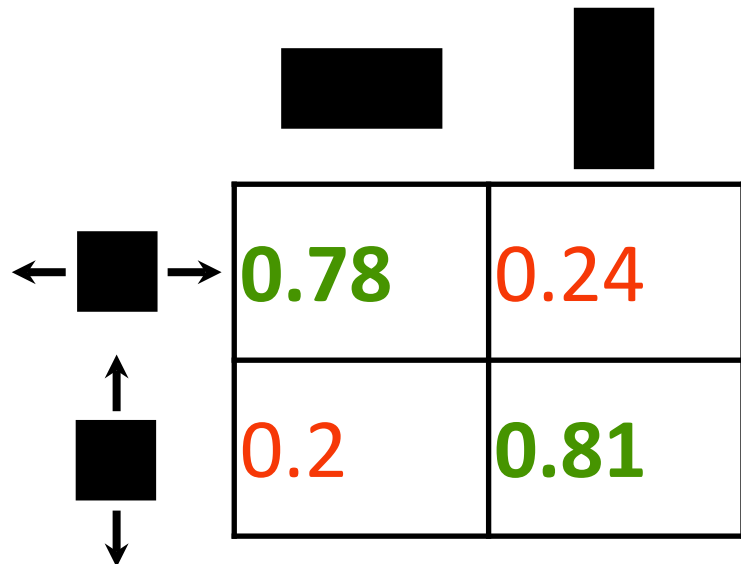
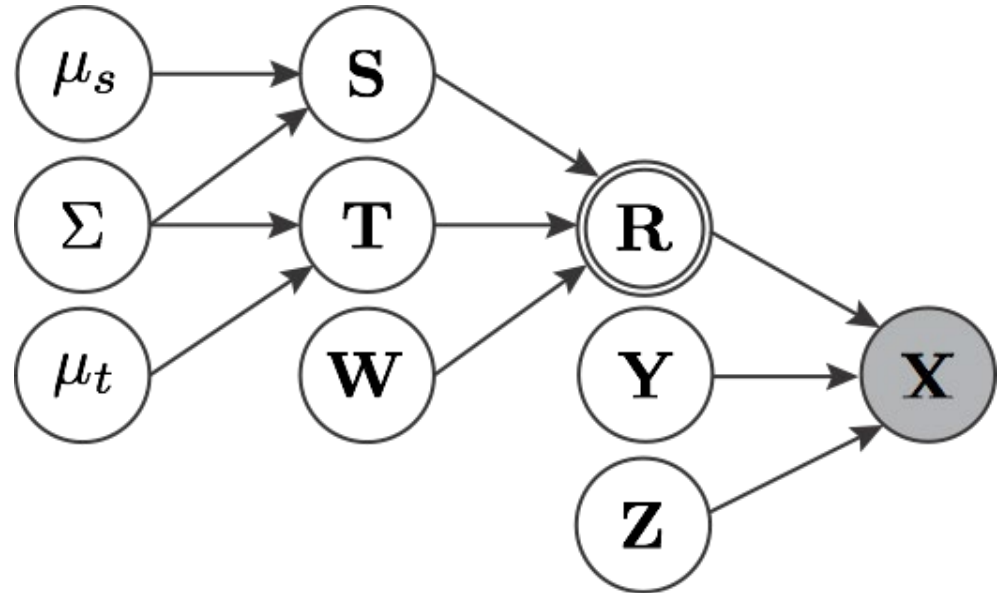
Learning feature transformation structure

$$s_{nk} \stackrel{iid}{\sim} N(\mu_s, \Sigma)$$

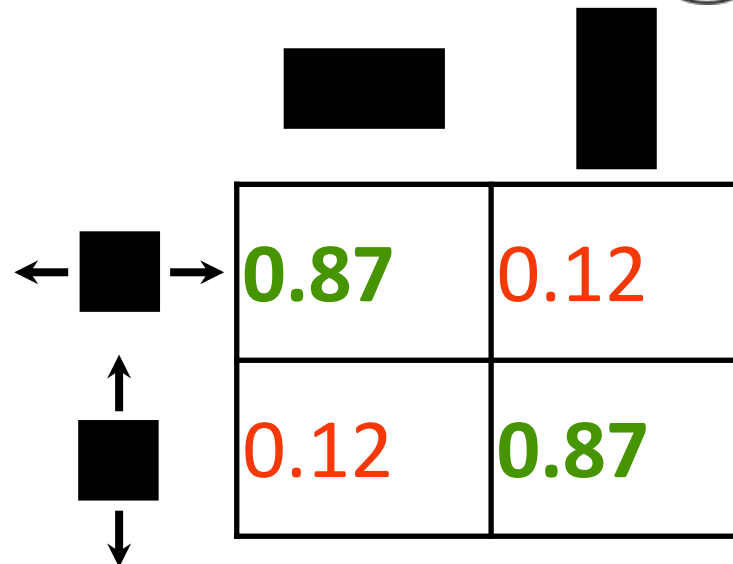
$$t_{nk} \stackrel{iid}{\sim} N(\mu_t, \Sigma)$$

$$w_{nk} \stackrel{iid}{\sim} \text{Bernoulli}(1/2)$$

$$r_{nk} = \begin{cases} s_{nk} & w_{nk} = 0 \\ t_{nk} & w_{nk} = 1 \end{cases}$$



Children generalization probabilities



Model generalization probabilities